

Optimal Carbon Tax with Dynamic Skill Heterogeneity

Alexandre Pavlov*

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Abstract

How should a government design a carbon tax when it is not only concerned by mitigating carbon emissions, but also income redistribution? This article extends a dynamic Mirrlees taxation model, typically used to characterize optimal labor wedges and savings wedges, to study the optimal commodity wedge between two goods, one of which is polluting. First, starting with the basic case where there are two non-polluting goods, I derive the optimal commodity wedge with non-separable preferences in a dynamic environment. Second, in the case where one of the goods is polluting and pollution damages both output and utility, I derive a modified optimal carbon tax whose output damages component remains Pigouvian and whose utility damages component is distorted by the labor wedge as well as the gap in complementarity in utility between each good and labor supply. Third, I derive a third-best commodity wedge when one of the other wedges (labor or savings) cannot be set optimally, mirroring the case when commodity wedges compensate for a suboptimal tax system, as well as a third best labor and savings wedge when the commodity wedge cannot be set optimally. In a third-best setting, the "free" wedges have to compensate for the suboptimal wedge. However, the direction of the adjustment of the "free" wedges depends on the structure of preferences, more specifically the size and sign of cross-derivatives.

Keywords: *Carbon tax, dynamic Mirrlees taxation, second-best environmental policy*

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*Université de Montréal: alexandre.pavlov@umontreal.ca

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1 Introduction

While there is a growing consensus on the existence of manmade climate change, there is still much disagreement on the appropriate policies to address it. Although carbon pricing is often lauded by experts as one of the most efficient tools to address climate change, claims that it may unfairly burden lower income households have stirred much public debate on carbon taxation in recent years in countries like France (*gilets jaunes* protests) and Canada (end of consumer carbon tax in early 2025).

However, integrated assessment models such as DICE, FUND, and PAGE which are used to inform carbon pricing policies (e.g. Environment and Climate Change Canada (2023), Interagency Working Group on Social Cost of Greenhouse Gases (2021), etc.) typically abstract from within-country inequalities. Indeed, these macroeconomic growth models with climate dynamics equations often view each country or region as a single unit and are therefore not designed to assess the distributional consequences of carbon pricing. How should a carbon tax then be designed to align with the government’s income redistribution goals?

Most research on the distributional aspects of carbon taxation takes an *ex post* perspective, asking whether existing carbon taxes are regressive and comparing the progressivity of different revenue recycling schemes. However, there is less knowledge on how income inequalities affects the optimal design of the carbon tax itself. Although policy documents such as the sixth IPCC assessment report stress the importance of considering within-country inequalities when designing carbon pricing, they stop short of providing concrete guidance (IPCC 2022).

There is already a longstanding literature on the optimal design of carbon taxes in the presence of distortionary income taxation, but it has focused on static representative-agent Ramsey models (Sandmo 1975; Bovenberg and de Mooij 1994; Goulder 1995). More recently, this literature has been extended to dynamic representative-agent Ramsey models (Barrage 2019). A common result in this literature is that the optimal carbon tax should be lower than the standard Pigouvian level, partially rationalizing why the price of carbon in most countries falls short of the value recommended by standard climate-economy models. For example, Barrage finds that the optimal carbon tax schedule lies 8-24% below the representative-agent Pigouvian benchmark in Golosov et al. (2014).

However, this result has been contested in Cremer et al. (1998) and Jacobs and de Mooij (2015), which use a static Mirrlees model to show that the optimal tax on a polluting good only differs from the Pigouvian level if preferences are non-separable between the polluting good and leisure. More recently, Douenne et al. (2026) and Douenne et al. (2025) have extended Barrage (2019) using the heterogeneous-agent Ramsey model developed in Werning (2007) and found with a calibrated model that the optimal carbon tax is only slightly lower than the Pigouvian level. Although this quantitative result may hint that the Mirrleesian irrelevance result extends to a dynamic environment, whether this is indeed true remains an open question.

Using a dynamic Mirrlees model of income taxation, I extend Jacobs and de Mooij 2015 to a dynamic environment where agents face uninsurable labor productivity shocks and their consumption generates an externality. In order to separate the role of commodity taxation and of externality correction, I first derive a formula for the optimal commodity tax in section 4. Then, I derive the main equations of a dynamic Mirrlees model with externalities in section 5. Afterwards, I consider in section 6 the case where a wedge is constrained suboptimal and see the implications of this third-best tax system before concluding in section 8.

2 Literature review

This paper sits at the intersection of three literatures that, until recently, evolved largely in parallel: the Ramsey tradition of second-best environmental taxation, the static Mirrleesian theory of externality correction under nonlinear income taxes, and the dynamic Mirrleesian theory of optimal income and capital taxation under private information. The contribution of this paper is most clearly seen as a bridge across all three, importing the full Mirrleesian apparatus of private information, nonlinear history-dependent taxation, and inverse-Euler intertemporal wedges into a dynamic climate-economy setting that the existing literature has so far treated only in Ramsey form. This section traces the route by which each of these literatures arrived at its current frontier, and shows why the frontier leaves precisely the gap this paper fills.

The starting point is Pigou's first-best: the optimal tax on a polluting good equals its marginal social damage. This benchmark survives unmodified in economies where the planner has access to lump-sum instruments, but it was not obvious whether it survived once distortionary taxes became part of the analysis. Sandmo (1975) established the first formal second-best result: in a Ramsey economy with linear labor taxes and a polluting good, the optimal tax formula is *additive* — marginal damages enter the formula for the polluting good alone, leaving the rest of the commodity tax system unchanged. But marginal damages are divided by the marginal cost of public funds (MCPF), which exceeds one when labor is distortionarily taxed. The implication: the optimal pollution tax lies strictly below marginal damages.

Bovenberg and de Mooij (1994) sharpened this into what became known as the "tax-interaction effect." In a revenue-neutral swap where environmental tax revenue is used to cut labor taxes, the environmental tax raises consumer prices, shrinks the real wage, and amplifies the pre-existing labor-supply distortion. Bovenberg and Goulder (1996) confirmed this numerically for U.S. carbon policy; Parry et al. (1999) quantified that the optimal pollution tax lies at 63–78% of marginal social damage. The policy implication was disheartening: inefficient tax systems should price carbon *less* aggressively, not more, because doing so compounds the inefficiency elsewhere. Goulder (1995)'s influential survey canonized this logic. The static Ramsey literature thus converged on a consensus that optimal pollution taxation lies strictly below Pigouvian, with the gap widening in the distortionariness of the broader tax system.

The bridge between static Ramsey second-best environmental taxation and the dynamic climate-economy models policymakers actually use was Golosov et al. (2014) (GHKT): under logarithmic utility, Cobb-Douglas production, and full capital depreciation each period, the optimal carbon tax as a share of GDP depends only on the discount rate, the expected damage elasticity, and the structure of carbon decay. Future paths of output, consumption, and population are irrelevant. This remarkable closed-form remains the benchmark for calibrated integrated assessment exercises, but the GHKT rate is a first-best representative-agent Pigouvian — it assumes away both distortionary taxation and heterogeneity.

Barrage (2019) extends GHKT to a dynamic Ramsey environment with distortionary labor and capital taxes and no lump-sum instruments. The optimal carbon tax schedule lies 8–24% below the GHKT benchmark, driven both by the division of utility damages by MCPF and by the rigidity of capital-tax instruments. Barrage also finds that recycling carbon revenue via capital-tax cuts yields much larger welfare gains than recycling via labor-tax cuts — a Chamley–Judd-flavored insight that carbon emissions destroy "natural capital." But this framework is representative-agent: inequality enters neither the planner's objective nor the constraints.

The most recent developments introduce heterogeneous agents into this dynamic Ramsey framework. Douenne et al. (2026) solves a complete-markets Ramsey problem with household heterogeneity, a DICE-calibrated climate module, and linear instruments (carbon, labor, capital) plus lump-sum transfers. In sharp contrast to Barrage (2019), the optimal carbon tax is essentially Pigouvian — inequality reduces the optimal rate by only about 4% — and revenue is optimally split roughly half-and-half between tax-rate reductions and transfer increases. Douenne et al. (2025) extends this to incomplete markets with uninsurable idiosyncratic labor-income risk à la Aiyagari, and shows that the Pigouvian result is quantitatively robust even under tight fiscal constraints: welfare gains come almost entirely from efficiency and environmental amenities, not redistribution or insurance. These two papers are, directly and correctly, the most important antecedents to the present work.

The empirical stakes of the distributional question are substantial regardless of whether the optimal rate is approximately Pigouvian. Känzig (2023) documents that carbon-pricing shocks in the EU Emissions Trading System impose consumption losses on low-income households that are roughly three times larger than on high-income households, with general-equilibrium adjustments in wages and employment accounting for two-thirds of the aggregate effect. Complementary contributions study third-best scenarios in which the instrument menu is restricted. van der Ploeg et al. (2025) find that when individualized transfers are unavailable and only a uniform carbon tax and uniform climate dividend can be used, high inequality aversion can push the optimal carbon tax *above* the Pigouvian level; Hänsel et al. (2022) derive analogous closed-form expressions under horizontal-equity constraints, showing that when the carbon tax must be uniform, the optimal marginal income tax becomes more progressive to offset heterogeneous

energy efficiency; Chiroleu-Assouline and Fodha (2014) show that regressive carbon taxes can be made Pareto-improving by coupling with more progressive labor taxation.

This arc narrows the gap between Ramsey and Pigouvian as instruments become more flexible, but remains within the Ramsey paradigm — linear instruments, observable types, period-by-period planning.

A parallel static literature had long argued that the Ramsey downward adjustment was not a robust feature of distortionary taxation but an artifact of the linear-instrument restriction. Cremer et al. (1998) showed that if individual consumption is observable, the nonlinear income tax is optimally set, and preferences are weakly separable, then *the carbon tax equals marginal damages* — the Sandmo additivity property survives, and the MCPF division disappears because the nonlinear income tax is itself optimized. Kaplow (2012) provided the cleanest version of the argument: any environmental-tax reform can be made *distribution-neutral* by adjusting the nonlinear income tax type-by-type, and under weak separability, a distribution-neutral move to the Pigouvian rate is Pareto-improving. The tax-interaction effect is therefore an artifact of conflating the environmental reform with an implicit change in the income tax.

Jacobs and de Mooij (2015) formalized the full result: when the nonlinear income tax is optimized, $MCPF = 1$ at the optimum, and the optimal pollution tax equals marginal damages — full stop — provided (i) the income tax is optimized, (ii) a lump-sum instrument exists, (iii) preferences are weakly separable in the externality, and (iv) preferences are type-homogeneous conditional on ability. When any of these conditions fails, the optimal pollution tax deviates from Pigouvian in signable directions: if the dirty good is a leisure complement (the Corlett–Hague case, substantiated for gasoline by West and Williams III (2007)), the carbon tax should exceed Pigouvian; if the dirty good is a necessity with non-homothetic Engel curves (Jacobs and van der Ploeg (2019)), it should fall below Pigouvian; and if high-ability households consume more of the dirty good, the redistributive role of commodity taxation can tilt the wedge in either direction (Christiansen (1984)).

The static Mirrleesian literature thus suggests that the Ramsey downward adjustment is an artifact of the linear-instrument restriction rather than a robust feature of distortionary taxation — but this claim had not been tested in a dynamic environment. Integrated assessment models, from which quantitative carbon-price recommendations are ultimately drawn, are intrinsically dynamic: carbon accumulates, skill draws evolve over the life cycle, and savings decisions link today’s choices to tomorrow’s consumption. Without a dynamic counterpart, the Mirrleesian Pigouvian result cannot discipline the time path of the optimal carbon tax.

In parallel, a separate literature developed the dynamic Mirrleesian theory of optimal taxation under private-information skill shocks, but abstracted from externalities. Golosov et al. (2003) established the *inverse Euler equation*: under separable preferences, the optimal allocation satisfies $1/u'(c_t) = (1/\beta R)\mathbb{E}_t[1/u'(c_{t+1})]$, which by Jensen’s inequality implies $u'(c_t) < \beta R\mathbb{E}_t u'(c_{t+1})$.

Agents' private Euler equations thus counsel over-saving relative to the constrained-efficient optimum, and the optimal intertemporal wedge is strictly positive whenever consumption risk is non-trivial. This is the fundamental break with the Chamley–Judd zero-capital-tax result: positive savings distortions are optimal because future consumption plays an asymmetric role in incentive compatibility — private saving relaxes the high-skill type's mimicking incentive.

Kocherlakota (2005) showed that this constrained-efficient allocation can be implemented by wealth taxes that are linear in wealth but nonlinear in labor-income history, with expected wealth tax equal to zero in every period. Albanesi and Sleet (2006) showed that for i.i.d. shocks, period-by-period history-independent taxes on current income and current wealth suffice. Kapička (2013) extended the first-order approach to persistent shocks and showed that persistence shrinks the average intertemporal wedge substantially, because persistent shocks leak information forward and reduce the marginal benefit of saving distortions. Farhi and Werning (2013) derived an AR(1)-like recursion for the labor wedge over the life cycle. Golosov et al. (2016) provided the tightest characterization: the optimal labor wedge has an ABCD decomposition into Frisch elasticity, elasticity of intertemporal substitution, redistributive, and hazard-rate components, and is U-shaped in productivity under calibrated skill processes. Within each period, Atkinson-Stiglitz uniform commodity taxation survives under weak separability; intertemporally, it fails because future consumption is a commodity whose MRS with current consumption is distorted by private future skills.

This apparatus provides the tools needed for a fully dynamic Mirrleesian treatment of the carbon tax — but contains no externality and no second commodity. The integration has not been made.

The picture that emerges is this. The Ramsey arc (Sandmo, Bovenberg–de Mooij, Parry et al., GHKT, Barrage, Douenne and coauthors) has pushed its dynamic heterogeneous-agent frontier to a quantitatively approximately Pigouvian result, but remains bound by linear instruments, observable types, and period-by-period planning. The static Mirrleesian externality literature (Cremer et al., Kaplow, Jacobs-de Mooij) has given us the analytical Pigouvian irrelevance result — but in a static, one-period environment. The dynamic Mirrleesian apparatus (Golosov et al., Kapička, Farhi-Werning, Golosov et al.) has given us the tools — private information, nonlinear history-dependent taxes, inverse-Euler intertemporal wedges — but no externality and no second commodity. The closest existing hybrid, Crampes and Ladoux (2024), treats energy-efficiency investment with nonlinear energy pricing in a two-period Mirrleesian setting, but does not address persistent skill shocks, capital taxation, or the full inverse-Euler structure.

A *fully* dynamic Mirrleesian treatment of carbon taxation — private-information skill shocks, nonlinear history-dependent wedges, inverse-Euler savings distortions, and a carbon externality in a multi-good consumption environment — has not, to my knowledge, appeared in the published literature. This paper fills that gap. The three contributions map directly onto the three strands: the optimal commodity-wedge derivation in Section 4 extends the dynamic

Mirrleesian apparatus from one to two goods; the corrective-wedge result in Section 5 extends Jacobs and de Mooij (2015)’s static irrelevance theorem to a dynamic private-information environment and, under non-separability, characterizes the signed deviations that Ramsey treatments cannot recover; the constrained-optimal analysis in Section 6 addresses the policy-economy concern of instrument restrictions in a fully Mirrleesian setting.

The relationship to Douenne and coauthors bears a final remark. Their results and the results here are *complementary*, not competing. Douenne et al. establish — quantitatively, in a heterogeneous-agent Ramsey setting — that the optimal carbon tax is close to Pigouvian; this paper establishes — analytically, in a heterogeneous-agent Mirrleesian setting — that this irrelevance result is a general feature of the interaction between a flexible redistribution instrument and an externality, provided weak separability holds. Douenne and coauthors measure the size of the efficiency-redistribution separation; this paper characterizes the preference conditions that make it exact, the non-separability channels that break it, and the third-best compensation patterns that take over when one instrument is locked down. The two approaches answer different questions with different tools. Read together, they describe a coherent picture of how optimal carbon taxation interacts with inequality in dynamic heterogeneous-agent economies.

3 Environment

A natural deviation from the standard dynamic Ramsey representative agent framework which explicitly models income heterogeneity and where taxation provides an economic benefit to agents is the dynamic Mirrlees heterogeneous agent framework developed in papers such as Golosov et al. (2003) and Golosov et al. (2016). I adapt this setup to an environmental context. In order to maintain focus on the effect of interhousehold inequality on the optimal design of the carbon tax, I abstract from other concerns such as international equity or intergenerational equity. As such, I consider a single generation and a single country.

The economy lasts T periods, with T being potentially infinite. Each agent consumes a vector of J goods $\mathbf{c}_t = (c_{1t}, \dots, c_{Jt})$, supplies labor l_t , and is exposed to aggregate pollution stock S_t . Lifetime preferences are

$$\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(\mathbf{c}_t, l_t, S_t) \right]$$

$\beta \in (0, 1)$ is the discount factor and good c_1 is the numéraire whose price is normalized to 1. The utility function U is time-separable, but otherwise allows non-separability between its arguments. A subset $\mathcal{J}_p$ of the J goods are polluting, with its complement $\mathcal{J}_c$ being "clean". Emissions from consuming the polluting goods contribute to a persistent pollutant stock S_t which evolves according to a function of current emissions as well as the past pollution stock.

$$S_t = S(S_0, S_1, \dots, S_{t-1}, Q_t)$$

This characterization of the evolution of the pollutant stock allows for current emissions to affect future pollutant accumulation, for example by having last period's pollution stock remain in the atmosphere after linear depreciation α .

$$S_t = \alpha S_{t-1} + Q_t$$

For generality, I do not restrict the form of the pollutant accumulation function. However, in order to be able to more easily rewrite the problem in a recursive form, I assume that emissions have no effect after s periods.

$$S_t = S(S_{t-s}, \dots, S_{t-1}, Q_t)$$

Aside from the presence of multiple goods whose subset generates pollution, the model follows the basic setup of the dynamic Mirrlees literature. At $t = 0$, agents draw their initial skill (type) θ_0 from a distribution $F_0(\theta)$. Subsequently, future skills are drawn from a Markov process $F_t(\theta|\theta_{t-1})$, meaning the likelihood of drawing a particular skill at time t depends on the skill drawn at time $t - 1$ but not on skills drawn previously. At period t , each agent knows their skill draws up to period t , which I denote by a skill history vector $\theta^t = (\theta_0, \dots, \theta_t)$. Let Θ^t be the set of all possible skill histories.

An agent of type θ_t who supplies l_t units of labor produces $y_t = \theta_t l_t$ units of output. Whereas total output is observed by everyone, both skill θ_t and labor supply l_t are private information. Moreover, consumption of all consumption goods is also publicly observable¹. Let $\mathbf{c}_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+^J$ and $y_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ be the agent's allocation of the consumption good vector and output, respectively. Let $\sigma^t(\theta^t) : \Theta^t \rightarrow \Theta^t$ be a function describing how an agent of type θ_t chooses to report their type and let Σ^t be all possible reporting strategies. Agents can use capital to transfer resources between periods, earning interest $R > 0$. The social welfare function is

$$\int_0^\infty \alpha(\theta) \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(\mathbf{c}_t, l_t, S_t) \right] dF_0(\theta)$$

$\alpha(\theta)$ is a non-negative Pareto weight on the welfare of an agent with initial skill θ . If the planner is utilitarian, $\alpha(\theta) = 1$ for all θ . Let U_{c_k} and U_l denote partial derivatives with respect to good c_k in the consumption vector $\mathbf{c}$ and l . Since $y = \theta l$, I can write $U(\mathbf{c}, l, S)$ as $U(\mathbf{c}, \frac{y}{\theta}, S)$. Let U_y and U_θ be the partial derivatives of utility with respect to y and θ .

Any admissible allocation must be incentive-compatible, meaning the agent must weakly prefer the truth-telling strategy over any other possible strategy. For all strategies and skill histories,

$$\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(\mathbf{c}_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(\mathbf{c}_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t \right) \middle| \theta \right]$$

¹This is a plausible assumption for some industries such as electricity and air travel. For a wider set of goods whose consumption cannot be as easily observed, the planner's optimum will be further constrained and the results in this paper no longer hold.

Moreover, for all periods, the resource constraints must hold.

$$\mathbb{E}_0[\mathbf{c}_t(\theta^t)\mathbf{I}_J|\theta] + k_{t+1} \leq \mathbb{E}_0[y_t(\theta^t)|\theta] + Rk_t$$

$\mathbf{c}_t(\theta^t)\mathbf{I}_J$ is a more compact way of writing $c_{1,t} + \dots + c_{J,t}$, with $\mathbf{I}_J$ being a J -sized identity vector. This set of constraints can be consolidated in a single intertemporal constraint.

$$\int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{\mathbf{c}_t(\theta^t)\mathbf{I}_J}{R} \middle| \theta \right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{y_t(\theta^t)}{R} \middle| \theta \right] dF_0(\theta)$$

Combining the planner's objective as well as the incentive-compatibility and resource-feasibility constraints yields the following planner's problem. For all strategies and skill histories,

$$\begin{aligned} & \max_{\{\mathbf{c}_t(\theta^t), y_t(\theta^t), S_t\}_{\theta^t \in \Theta^t, t=0, \dots, T}} \int_0^\infty \left(\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(\mathbf{c}_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t} \right) \middle| \theta \right] \right) dF_0(\theta) \\ & \text{s.t.} \\ & \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(\mathbf{c}_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t} \right) \middle| \theta \right] \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(\mathbf{c}_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t} \right) \middle| \theta \right] \\ & \int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{\mathbf{c}_t(\theta^t)\mathbf{I}_J}{R} \middle| \theta \right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{y_t(\theta^t)}{R} \middle| \theta \right] dF_0(\theta) \\ & S_t = S(S_{t-s}, \dots, S_{t-1}, Q_t) \end{aligned}$$

Now that we have setup the problem, we are ready to characterize the optimal joint tax system, which now includes a corrective wedge.

4 Optimal dynamic non-linear commodity wedge

Since a general formula for the optimal commodity wedge in a dynamic Mirrlees environment with non-separable preferences has not yet been derived in the literature, I start by deriving it in an economy with multiple goods but no pollution. For an agent with history θ^t , I define labor distortions, savings distortions, and commodity distortions as follows.

$$\begin{aligned}
1 - \tau_t^y(\theta^t) &\equiv -\frac{U_l\left(\mathbf{c}_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right)}{\theta_t U_c\left(\mathbf{c}_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right)} \\
1 - \tau_t^s(\theta^t) &\equiv \frac{1}{\beta R} \frac{U_c\left(\mathbf{c}_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right)}{\mathbb{E}_t \left[U_c\left(\mathbf{c}_{t+1}(\theta^{t+1}), \frac{y_{t+1}(\theta^{t+1})}{\theta_{t+1}}\right) \right]} \\
1 + \tau_t^{c_j}(\theta^t) &\equiv \frac{U_{c_j}\left(\mathbf{c}_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right)}{U_c\left(\mathbf{c}_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}\right)} \quad j = 2, \dots, J
\end{aligned}$$

To keep notation light, I write u_c instead of u_{c_1} . Moreover, I define the following elasticities

- $\epsilon_t(\theta) \equiv \frac{U_{ll,t}(\theta)l_t(\theta)}{U_{l,t}(\theta)}$: the inverse Frisch elasticity of labor supply
- $\sigma_t(\theta) \equiv -\frac{U_{cc,t}(\theta)c_t(\theta)}{U_{c,t}(\theta)}$: the elasticity of intertemporal substitution of the numéraire
- $\gamma_{k,t}(\theta) \equiv \frac{U_{c_k l,t}(\theta)l_t(\theta)}{U_{c_k,t}(\theta)}$: the level of complementarity between good c_k and labor supply

The problem can be written recursively, using the results in Kapička (2013)².

$$\begin{aligned}
V_t(\hat{w}, \hat{w}_2, \theta_-) &= \min_{\mathbf{c}, y, u, w, w_2} \int_0^\infty \left[\mathbf{c}(\theta) \mathbf{I}_J - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta) \right] f_t(\theta | \theta_-) d\theta \\
&\quad \text{s.t.} \\
\dot{u}(\theta) &= U_\theta \left(\mathbf{c}(\theta), \frac{y(\theta)}{\theta} \right) + \beta w_2(\theta) \\
\hat{w} &= \int_0^\infty u(\theta) f_t(\theta | \theta_-) d\theta \\
\hat{w}_2 &= \int_0^\infty u(\theta) f_{2,t}(\theta | \theta_-) d\theta \\
u(\theta) &= U \left(\mathbf{c}(\theta), \frac{y(\theta)}{\theta} \right) + \beta w(\theta)
\end{aligned}$$

Solving the above problem, I find that the structure of the optimal labor income tax and of the optimal capital income tax is unchanged from Golosov et al. 2016. However, due to the presence of multiple goods, there are commodity wedges between the numéraire and the other goods which is non-zero when utility is non-separable between labor and the non-numéraire.

Result 4.0.1 (Optimal commodity wedge).

The optimal commodity wedge between a good j and the numéraire can be expressed as

$$\frac{\tau^{c_j}}{1 + \tau^{c_j}} = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \left(\frac{\tau_{t-1}^{c_j}}{1 + \tau_{t-1}^{c_j}} \right) D_t(\theta)$$

$$\begin{aligned}
A_t(\theta) &= \gamma_{1,t}(\theta) - \gamma_{j,t}(\theta) \\
B_t(\theta) &= \frac{1 - F_t(\theta)}{\theta f_t(\theta)} \\
C_t(\theta) &= \int_\theta^\infty \exp \left[\int_\theta^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
D_t(\theta) &= \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_\theta^\infty \exp \left[- \int_\theta^x \gamma_{1,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}
\end{aligned}$$

Proof. See appendix A.1. □

²Kapicka considers that consumption comes from an arbitrary convex and compact set, which means it is not restricted to a scalar and can very well be a vector of multiple consumption goods.

The optimal commodity wedge in (4.01) addresses the same fundamental tension as the optimal labor wedge in Golosov et al. (2016): how to redistribute and insure against shocks to low-skilled types subject to the incentive-compatibility constraint that high-skilled types not mimic them. The labor wedge addresses this tension directly, by raising the marginal cost of labor at type θ so that the bundle intended for low types becomes unattractive to mimickers. The commodity wedge addresses the same tension *indirectly*, by exploiting a consequence of non-separable preferences: when $U_{c_k l} \neq 0$, the marginal utility of each consumption good depends on labor supply, so the relative valuation of two goods depends on how hard one works. A high-type pretending to be a low type values labor-complement goods less than the true low type would, and leisure-complement goods more. The planner exploits this asymmetry by tilting the low-type bundle towards goods that mimickers value less (a subsidy on labor-complements) and away from goods that mimickers value more (a tax on leisure-complements). This makes the bundle less attractive to mimickers without making it less attractive to true low types, relaxing the incentive constraint without further distorting the labor margin.

The leverage available to this instrument is what the term $A_t(\theta) = \gamma_{1,t}(\theta) - \gamma_{j,t}(\theta)$ measures: the gap in labor-complementarity between the numéraire and good j . When the two goods enter labor's cross-derivative symmetrically ($\gamma_1 = \gamma_j$), the mimicker and the true low type value the consumption mix identically, the commodity instrument has no informational handle, and the wedge collapses to zero. When good j is less labor-complementary than the numéraire, $A_t > 0$ delivers a positive tax on j ; when more labor-complementary, a subsidy. This is the dynamic Mirrleesian generalisation of the Corlett–Hague rule presented in Christiansen (1984). Under separability between labor and the consumption goods, the commodity wedge vanishes and the planner is left with the labor wedge as the solve instrument, which is the dynamic Atkinson–Stiglitz benchmark of Atkinson and Stiglitz (1976) and Golosov et al. (2003). Like the labor wedge, the commodity wedge aims to relax the incentive constraint at type θ , the benefit of which accrues to the mass $1 - F_t(\theta)$ of higher types while the cost is borne at the productivity-weighted local mass $\theta f_t(\theta)$.

Once the incentive constraint is relaxed, the resources the planner can extract from higher types must be valued against the social cost of taking them. Because the destination of extracted resources does not depend on which instrument the planner used to extract them, C_t is the same in both formulas. The dynamic term $D_t^{c_j}(\theta)$ inherits an analogous parallel: as in Golosov et al. (2016), it captures how today's wedge is used to back out incentives for past type-revelation, the likelihood-ratio integral $\int_{\theta}^{\infty} \exp\left[-\int_{\theta}^x \gamma_{1,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x) dx$ measuring how informative today's θ is about θ_{t-1} , while the ratios A_t/A_{t-1} and $U_{c,t}(\theta)/U_{c,t-1}$ encode the same principle as for the labor wedge: it is cheaper to provide incentives in states of high marginal utility and low instrument-responsiveness. The persistence factor $\tau_{t-1}^{c_j}/(1+\tau_{t-1}^{c_j})$ means that commodity wedges have their own dynamics, propagated through their own lags rather than the labor wedge's.

The story is therefore one of parallel substitutes. The labor wedge and the commodity wedge

are alternative instruments for the same job: relaxing the incentive constraint at low types so as to enable redistribution. They share the hazard-ratio cost–benefit trade-off (B), the social-value-of-redistribution machinery (C), and the dynamic-incentive structure (D) and differ only in the channel through which they extract incentive information. Three implications follow. First, the Atkinson-Stiglitz uniform-commodity-taxation theorem is recovered as a corollary in the present dynamic environment: separability sends $\gamma_1, \gamma_j \rightarrow 0$, hence $A_t \rightarrow 0$, which collapses the intratemporal component and, via A_t in D_t together with $\tau_{t-1}^{c_j} = 0$, the intertemporal component as well, leaving the labor wedge as the only active instrument. Second, under constant elasticities the intratemporal commodity wedge is exactly proportional to the intratemporal labor wedge across (t, θ) , with constant of proportionality $(\gamma_1 - \gamma_j)/(1 + \varepsilon)$. Hence, the shape of the commodity wedge is determined by the shape of the labor wedge. Third, history-dependence is commodity-specific: even with constant elasticities, D_t multiplies $\tau_{t-1}^{c_j}/(1 + \tau_{t-1}^{c_j})$, so the commodity wedge carries its own intertemporal persistence, a feature absent from the static multi-good Mirrleesian models.

5 Optimal dynamic non-linear wedges with externalities

Now that we have characterized optimal commodity wedges in a dynamic Mirrleesian economy, we are ready to introduce externalities. To simplify notation, let us specialize to $J = 2$ and consider two goods denoted by c and q , where the first one is the clean numéraire and the second one is polluting. The global incentive constraints in the planner's problem can be substituted using the results in Kapička (2013) to yield the following recursive problem.

$$\begin{aligned}
V_t(\hat{w}, \hat{w}_2, \theta_-, S_-) &= \min_{c, q, y, u, w, w_2, S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, S) \right] f_t(\theta|\theta_-) d\theta \\
&\quad \text{s.t.} \\
\dot{u}(\theta, S) &= U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w_2(\theta) \\
\hat{w} &= \int_0^\infty u(\theta) f_t(\theta|\theta_-) d\theta \\
\hat{w}_2 &= \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta \\
u(\theta, S) &= U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w(\theta) \\
S &= S \left(\dots, \int q(\theta) f(\theta) d\theta \right)
\end{aligned}$$

Solving this problem, the structure of income taxes is once again unchanged. However, the

commodity wedge now features what amounts to a modified Pigouvian term where marginal damages of pollution to utility are adjusted for the planner's fiscal concerns.

Result 5.0.1 (Optimal corrective wedge).

The optimal corrective wedge between the "clean" good and the "polluting" good is

$$\frac{\tau_t^q - SCC_t}{1 + \tau_t^q} = A_t(\theta)B_t(\theta)C_t(\theta) + \beta R \left(\frac{\tau_{t-1}^q - SCC_{t-1}}{1 + \tau_{t-1}^q} \right) D_t(\theta)$$

where

$$SCC_t = - \int \left(\sum_{j=0}^s \frac{1}{R^j} \left[1 + \left(\frac{\tau_j^y}{1 - \tau_j^y} \right) \frac{\xi_j - \gamma_{c,j}}{1 + \epsilon_j - \gamma_{c,j}} \right] \frac{U_{S,j}}{U_{c,j}} S_{s-j} \right) f_t$$

$$A_t(\theta) = \gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)$$

$$B_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$C_t(\theta) = \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)}$$

$$D_t(\theta) = \frac{A_t(\theta) U_{c,t}(\theta)}{A_{t-1} U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{1,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}$$

Proof. See appendix B.1. □

The optimal corrective wedge in (5.0.1) decomposes into two economically distinct components. We have a familiar Mirrleesian component, which takes the same shape as in the pollution case, and a new corrective component, which features a modified Pigouvian term SCC_t . However, the commodity wedge now operates on the deviation $\tau_t^q - SCC_t$ rather than on τ_t^q itself. The SCC_t is itself worth dwelling on. It is the present-discounted sum, over the s periods through which carbon emissions raise the pollution stock, of marginal pollution damages $-(U_{S,j}/U_{c,j})S_{s-j,j}$. However, each period's damage is multiplied by a fiscal adjustment factor $1 + (\tau_j^y/(1 - \tau_j^y))(\xi_j - \gamma_{c,j})/(1 + \epsilon_j - \gamma_{c,j})$. This factor equals one, recovering the standard Pigouvian social cost of carbon, whenever the labor wedge is absent ($\tau_j^y = 0$) or whenever the pollution stock interacts with labor identically to the clean good ($\xi_j = \gamma_{c,j}$). When both $\tau_j^y \neq 0$ and $\xi_j \neq \gamma_{c,j}$, the planner incorporates the fiscal externality of carbon abatement on labor supply: through the cross-derivative U_{lS} , a cleaner environment shifts the marginal utility of labor in a way that interacts with the existing labor distortion. If $\xi_j > \gamma_{c,j}$, abatement provides a fiscal bonus, partially undoing the labor wedge, and the SCC exceeds the pure Pigouvian level. If $\xi_j < \gamma_{c,j}$,

abatement aggravates the labor distortion and the SCC falls below the Pigouvian level. This is the "virtual subsidy" channel of Jacobs and de Mooij (2015), generalised here to a dynamic, stock-pollutant environment.

When the Mirrleesian motive disappears, either because goods are equally complementary with labor or because preferences are separable between consumption and labor, the planner sets $\tau_t^q = \text{SCC}_t$ exactly. That is, unlike in a dynamic Ramsey environment such as Barrage (2019) or Douenne et al. (2026), the presence of positive labor and capital taxation does not imply a deviation from the standard Pigouvian rule unless there is non-separability between labor supply and the other variables.

The Mirrleesian component inherits the entire interpretation of Result 4.01 verbatim. $A_t(\theta) = \gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)$ measures the leverage available to the planner from using the consumption mix between c and q to discriminate high-type mimickers from genuine low types: under non-separability the relative valuation of clean and dirty goods depends on labor supply, and the planner exploits the resulting asymmetry to relax the incentive constraint at low types. $B_t(\theta)$ is the same hazard-ratio cost-benefit trade-off, $C_t(\theta)$ is the same redistributive value of extracted resources, and $D_t(\theta)$ is the same intertemporal incentive term, now propagated through the lagged-deviation factor $(\tau_{t-1}^q - \text{SCC}_{t-1})/(1 + \tau_{t-1}^q)$ rather than through the lagged $\tau_{t-1}^q/(1 + \tau_{t-1}^q)$ alone.

Three implications follow. First, the textbook dynamic Pigouvian carbon tax is recovered only on a knife-edge: separable preferences (which silence the Mirrleesian wedge) combined with either $\tau_j^y = 0$ or $\xi_j = \gamma_{c,j}$ in every period (which silences the SCC adjustment). Outside this set, the optimal carbon tax deviates from textbook Pigou in two structurally distinct ways, through the SCC fiscal adjustment and through the Mirrleesian commodity wedge, which can reinforce or offset one another depending on the signs of $\gamma_c - \gamma_q$ and $\xi - \gamma_c$. Second, the dynamic structure of the wedge places the persistence of past deviations $\tau_{t-1}^q - \text{SCC}_{t-1}$, not of past carbon taxes themselves, into the intertemporal incentive role. The Pigouvian component is forward-looking (a present-discounted sum of future damages); the Mirrleesian component is backward-looking (a persistence of past redistributive wedges); the two motives meet only in the period- t optimality condition. Third, the framework cleanly separates corrective and redistributive motives in the carbon tax. The SCC handles the externality, including its fiscal interaction with the labor wedge while the Mirrleesian portion handles redistribution and insurance. Both depend on labor-tax policy through τ^y (the SCC adjustment factor) and on the structure of preferences through ε , γ , and ξ , but they remain conceptually separable instruments jointly optimised by the planner.

6 Constrained-optimal dynamic non-linear wedges with externalities

The second-best results so far show that the structure of the rest of the tax system is unaffected by the presence of the externality, suggesting there might not necessarily be a gain from using proceeds from a carbon tax to fund reductions in other taxes, as is often advocated in the policy debate. However, the second-best results also implicitly assume that the planner jointly optimizes all tax instruments, which is not necessarily the case in practice. If the planner has to contend with wedge constraints, perhaps due to feasibility or political-economy reasons, how should they optimally adjust the rest of the tax system?

Mathematically, this amounts to solving the planner's problem with an added constraint that the wedge must equal an exogenous schedule, where it can only optimize with respect to the remaining free wedges. I consider two possible cases: (i) the commodity wedge is fixed and (ii) the labor and savings wedges are fixed. In the first case, we would like to know how the labor and savings wedges would be optimally set if the commodity wedge were suboptimal. For instance, in the fully separable case where the commodity wedge is solely composed of a corrective component, this can be interpreted as a situation where the polluting good is suboptimally taxed under the Pigouvian level. In the second case, we would like to know how the commodity wedge would be set if the other wedges were suboptimal. Note that I do not restrict the form of the exogenous schedule and it may indeed be non-linear. Hence, when I say that a wedge is underdistorted or overdistorted, I am talking about the wedge for individual θ .

6.1 Constrained-optimal commodity wedge (suboptimal labor wedge)

Proposition 6.1.1 (Third best commodity wedge (constrained labor wedge)).

The optimal commodity wedge when the labor wedge is constrained to $\bar{\tau}^y(\theta)$ is

$$\frac{\tau^q - \mu S_q + \frac{\nu^y}{f}(1 - \bar{\tau}^y)\eta_q}{1 + \tau^q} = A_t(\theta)B_t(\theta)C_t(\theta) + \beta R \left(\frac{\tau_{t-1}^q - \mu S_{q,t-1} + \frac{\nu_{t-1}^y}{f_{t-1}}(1 - \bar{\tau}_{t-1}^y)\eta_{q,t-1}}{1 + \tau_{t-1}^q} \right) D_t(\theta) +$$

$$\frac{\nu_t^y}{f_t}(1 - \bar{\tau}_t^y)\eta_{c,t} - \beta R \frac{\nu_{t-1}^y}{f_{t-1}}(1 - \bar{\tau}_{t-1}^y)\eta_{c,t-1} D_t(\theta)$$

where

$$\begin{aligned}
A_t(\theta) &= \gamma_{c,t}(\theta) - \gamma_{q,t}(\theta) \\
B_t(\theta) &= \frac{1 - F_t(\theta)}{\theta f_t(\theta)} \\
C_t(\theta) &= \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] \left(1 - \frac{\nu_t^y}{f_t(x)} (1 - \bar{\tau}_t^y(x)) \eta_{c,t}(x) - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x) \right) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
D_t(\theta) &= \frac{A_t(\theta)}{A_{t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}
\end{aligned}$$

and η_x is the semi-elasticity of the labor wedge with respect to variable x .

Proof. See appendix C.1 □

REWRITE INTERPRETATION

6.2 Constrained-optimal commodity wedge (suboptimal savings wedge)

Suppose the savings wedge is constrained at $\bar{\tau}^s(\theta)$. Define $g(w, w_2, \theta, Q) \equiv \mathbb{E}_t[U_{c,t+1}]$. Future consumption is affected by promised utility and future promised utility as well as the current skill shock (due to persistence) and aggregate polluting good consumption which will affect future damages. How does this change the optimal commodity wedge?

TBA

6.3 Constrained-optimal labor and savings wedges

Conversely, suppose the commodity wedge is constrained to $\bar{\tau}^g(\theta)$ and the labor and savings wedges are free. For example suppose the commodity wedge cannot be set optimally due to practical considerations or political feasibility.

Proposition 6.3.1 (Third-best labor wedge (constrained commodity wedge)).
The optimal labor wedge when the commodity wedge is constrained to $\bar{\tau}^g(\theta)$ is

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t(\theta)B_t(\theta)C_t(\theta) + \left[\left(\frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \right) + \nu_{t-1}^g \Gamma_{q,t-1}^g \right] D_t(\theta) + \nu_t^g \Gamma_{l,t}^g$$

where

$$\begin{aligned} A_t(\theta) &\equiv (1 + \epsilon_t(\theta) - \gamma_t(\theta)) \\ B_t(\theta) &\equiv \frac{1 - F_t(\theta)}{\theta_t f_t(\theta)} \\ C_t(\theta) &\equiv \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 + \nu_t^g \Gamma_{q,t}^g - \lambda_{1,t} U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\ D_t(\theta) &\equiv \frac{A_t(\theta)}{A_{t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)} \end{aligned}$$

and

$$\begin{aligned} \Gamma_{l,t}^g &\equiv \frac{U_{cq}}{U_{cf}} - \frac{U_{ql}}{U_{lf}} - (1 + \bar{\tau}^g) \left(\frac{U_{cc}}{U_{cf}} - \frac{U_{cl}}{U_{lf}} \right) \\ \Gamma_{q,t}^g &\equiv \frac{U_{cq}}{U_{cf}} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_{cf}} \end{aligned}$$

Proof. See appendix C.3. □

The third-best labor wedge above is the labor-wedge analog of the constrained-commodity-wedge result in section 6.2: when the planner cannot freely choose one wedge, the other absorbs the constraint's shadow value, and the absorption shows up at three distinct locations in the ABCD form. The unconstrained dynamic-Mirrleesian labor wedge of Golosov et al. (2016) is recovered as a smooth $\nu^g \rightarrow 0$ limit; all three of its structural components $A_t = 1 + \epsilon_t - \gamma_t$, $B_t = (1 - F_t)/(\theta_t f_t)$, and D_t are unchanged. The Frisch-augmented labor-margin elasticity still

sets the cost of distorting work; the inverse-hazard ratio still sets the cost–benefit trade-off; the persistence kernel still propagates yesterday’s wedge through the same likelihood-ratio integral. What changes is exclusively how the commodity-wedge constraint colors the redistributive valuation C_t , the lagged-wedge object inside the persistence bracket, and a single new additive term $\nu_t^g \Gamma_{l,t}^g$ on the RHS.

The integrand of C_t picks up the augmentation $1 + \nu^g \Gamma_q^g$ in place of the unconstrained 1 inside the $1 - \lambda_1 U_c$ kernel. The new piece $\Gamma_q^g = U_{cq}/(U_c f) - (1 + \bar{\tau}^g) U_{cc}/(U_c f)$ measures how a marginal increase in c shifts the relative-price ratio U_q/U_c that the commodity constraint pins down. Under fully separable preferences with $U_{cc} < 0$ and $\bar{\tau}^g > -1$, $\Gamma_q^g = -(1 + \bar{\tau}^g) U_{cc}/(U_c f) > 0$, so when the commodity wedge is under-distorted ($\nu^g > 0$) the entire redistributive valuation C_t is rescaled upward. The reading is direct: when the consumption-mix lever cannot be fine-tuned, the planner accepts a higher social cost of redistribution because she cannot reach the high-type’s mimicker bundle through the commodity instrument. What was a free-margin redistribution becomes a more costly labor-margin redistribution. Beyond separability, Γ_q^g depends on how U_{cq} compares to $(1 + \bar{\tau}^g) U_{cc}$. Strong Edgeworth complementarity between c and q ($U_{cq} > 0$) can flip Γ_q^g negative, in which case binding under-distortion of the commodity wedge actually shrinks $|C_t|$ and the Mirrleesian motive is muted.

The persistence channel inherits the same Γ_q^g structure but operates one period earlier. The lagged bracket $[\tau_{t-1}^y/(1 - \tau_{t-1}^y) + \nu_{t-1}^g \Gamma_{q,t-1}^g] D_t$ propagates yesterday’s labor wedge augmented by yesterday’s commodity-constraint shadow. The interpretation is the time-shifted twin of the constrained-commodity case in section 6.2: yesterday’s constraint failure is carried forward through the same intertemporal-incentive kernel that propagates yesterday’s redistribution, so the planner’s dynamic accommodation of past commodity-wedge constraints and her incentive role for past labor wedges travel together through D_t and cannot be separated at the optimum.

The genuinely new piece is the contemporaneous additive term $\nu_t^g \Gamma_{l,t}^g$ on the RHS, with no structural analog in Result 4.0.1 or 5.0.1’s recursion. Where Γ_q^g measures the c -channel response of the commodity wedge, $\Gamma_l^g = U_{cq}/(U_c f) - U_{ql}/(U_l f) - (1 + \bar{\tau}^g)[U_{cc}/(U_c f) - U_{cl}/(U_l f)]$ measures the cross-margin response that links labor supply to the consumption composition. Under fully separable preferences $\Gamma_l^g = \Gamma_q^g$, and the two corrections simply add: when the commodity wedge is too low, the labor wedge increases by twice the separable-benchmark Γ_q^g correction, once through the modified C_t , once through the standalone term. The planner over-distorts labor as compensation for the under-distorted consumption mix, exactly the policy implication Jacobs and de Mooij (2015) flags in static models, here with a fully dynamic structure. Under non-separable preferences the two Γ objects move independently: Γ_l^g picks up U_{cl}/U_l and U_{ql}/U_l terms that the consumption-margin Γ_q^g does not see, so the policy direction is no longer mechanical and: a commodity wedge that is too low for type θ can require either over- or under-distorting the labor margin, depending on whether Γ_l^g and Γ_q^g point in the same direction.

Three implications follow. First, the labor-wedge formula inherits the entire intertemporal

architecture of Golosov et al. (2016). The commodity-wedge constraint enters as additive corrections only, not as a structural transformation. Second, the constrained labor and commodity problems are not perfectly symmetric. Third, even if we assume separable preferences, the policy prescription is ambiguous as it depends on the structure of preferences. When commodity wedges cannot be set optimally, the planner should respond either by over-distorting labor supply to compensate, or alternatively by under-distorting it.

Proposition 6.3.2 (Third-best savings wedge (constrained commodity wedge)).
The optimal savings wedge when the commodity wedge is constrained to $\bar{\tau}^g(\theta)$ is

Proof. See appendix C.4 (TBA). □

7 Simulations

TBA

8 Conclusion

Using a dynamic Mirrlees framework, which features an explicit motive for income taxation, extended to feature consumption externalities, I find that the optimal pollution tax is only adjusted for inequality when there are non-separabilities in preferences between consumption, labor, and pollution. Otherwise, taxes are Pigouvian even in the presence of labor and capital income taxation, which shows that the dynamic Ramsey result that tax distortions justify deviating from Pigouvian taxation does not necessarily hold. Moreover, the optimal formulae for labor and savings wedges remain unchanged, suggesting that a properly designed commodity wedge is fully capable of internalizing the externality while respecting the social planner's redistributive goals.

When considering a third-best setting where one of the wedges is fixed at an arbitrary exogenous schedule, I find that the other wedges must be adjusted to compensate for the (potential) suboptimality of the fixed wedge. However, the direction of the adjustment depends on the structure of preferences. For example, when preferences are fully separable, underdistorting labor supply requires overdistorting commodity choices. If preferences are non-separable and some cross-elasticities are sufficiently large, the direction of the adjustment might flip.

There is much room for further study. First, this paper focuses on wedges rather than tax implementation, so an interesting avenue of research may be assessing how the optimal commodity wedge may be replicated in practice. Second, the paper features a single generation of potentially infinitely-lived individuals although climate change is in practice very much an intergenerational problem. Therefore, extending a heterogeneous framework to feature overlapping generations is another promising next step. Outside of the environmental context, since the optimal commodity wedge is a time-varying object in this dynamic model, one might also wonder if and how age-dependent commodity taxes may be used in a fiscal mix. Although age-dependent commodity taxation may at first glance appear a fanciful notion, positive age discrimination in the price of certain goods and services for young or old consumers or indeed discounts for low-income consumers already somewhat acts as implementing a time-varying income-based commodity wedge³.

³Of course, whether this is done for social or economic reasons is another question. However, the point is that it *can* be done. Future work could evaluate the possible deliberate use of age-dependent income-dependent commodity wedges to further efficiency or equity objectives.

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A Optimal wedge derivation without externalities

A.1 Commodity wedge

Because $U(\mathbf{c}, l)$ can be written as $U(\mathbf{c}, y/\theta)$, I can write U_θ as $-U_l l/\theta$. The Hamiltonian to this problem is as follows.

$$\begin{aligned}\mathcal{H} = & \left[\mathbf{c}\mathbf{I}_J - \theta l + \frac{1}{R}V_{t+1}(w, w_2, \theta) \right] f \\ & + \psi \left[-U_l(\mathbf{c}, y/\theta) \frac{l}{\theta} + \beta w_2 \right] \\ & - \lambda_1 u(\theta) f_t \\ & + \lambda_2 u(\theta) f_2 \\ & + \phi [u - U(\mathbf{c}, y/\theta) - \beta w]\end{aligned}$$

This yields the following first-order conditions.

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial c} : f - \psi U_c \frac{l}{\theta} &= \phi U_c \\ \frac{\partial \mathcal{H}}{\partial c_j} : f - \psi U_{c_j} \frac{l}{\theta} &= \phi U_{c_j} \quad j = 2, \dots, J \\ \frac{\partial \mathcal{H}}{\partial l} : -\theta f - \phi U_l &= \frac{1}{\theta} \psi [U_{ll} l + U_l] \\ \frac{\partial \mathcal{H}}{\partial u} : -\lambda_1 f + \lambda_2 f_2 + \phi &= \dot{\psi} \\ \frac{\partial \mathcal{H}}{\partial w} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f &= \phi \beta \\ \frac{\partial \mathcal{H}}{\partial w_2} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f &= -\psi \beta\end{aligned}$$

Because the first-order conditions are unchanged, aside from the presence of $\frac{\partial \mathcal{H}}{\partial c_j}$ which is not involved in the derivation of non-commodity wedges, the steps to derive the expression for the optimal labor and savings income wedges are identical to Golosov et al. (2016). Hence, I refer to this paper for details.

There are different equivalent ways to derive and express the commodity wedge between good j and the numéraire good, but I will seek to express it in the familiar labor wedge ABCD form. First, because the first-order conditions for each good share the same ϕ ,

$$\frac{f_t}{U_{c_j}} - \psi \frac{U_{c_j,l}}{U_{c_j}} \frac{l}{\theta} = \frac{f_t}{U_c} - \psi \frac{U_{c,l}}{U_c} \frac{l}{\theta}$$

If we cross-multiply by U_{c_j} and U_c as well as use $\gamma_k \equiv \frac{U_{c_k,l}}{U_{c_k}}$,

$$U_c \left(f - \frac{\psi}{\theta} \gamma_j U_{c_j} \right) = U_{c_j} \left(f - \frac{\psi}{\theta} \gamma_1 U_c \right)$$

Then, if we divide by $U_c f$ and rearrange terms,

$$\frac{U_c}{U_{c_j}} - 1 = \frac{\psi U_c}{\theta f} (\gamma_j - \gamma_1)$$

Note that, by definition, $1 + \tau^{c_j} = \frac{U_{c_j}}{U_c}$, so $\frac{U_c}{U_{c_j}} - 1 = -\frac{\tau^{c_j}}{1 + \tau^{c_j}}$. Multiplying through by -1 ,

$$\frac{\tau^{c_j}}{1 + \tau^{c_j}} = \frac{\psi U_c}{\theta f} (\gamma_1 - \gamma_j)$$

We can find the expression for ψ by isolating it in $\frac{\partial \mathcal{H}}{\partial w_2}$.

$$\psi = -\frac{1}{\beta R} \frac{\partial V_{t+1}}{\partial w_2} f$$

Likewise, we can find another expression for ψ .

$$\psi = \left(\frac{\tau^{c_j}}{1 - \tau^{c_j}} \right) \frac{\theta f}{U_c} \frac{1}{\gamma_1 - \gamma_j}$$

Bringing these two expressions together,

$$-\frac{1}{\beta R} \frac{\partial V_{t+1}}{\partial w_2} = \left(\frac{\tau^{c_j}}{1 - \tau^{c_j}} \right) \frac{\theta}{U_c} \frac{1}{\gamma_1 - \gamma_j}$$

To find the expression for $\psi(\theta)$, we can integrate $\frac{\partial \mathcal{H}}{\partial u}$ from θ to ∞ .

$$\dot{\psi} = -\lambda_1 \bar{\alpha}_t f + \lambda_2 f_2 + \frac{f}{U_c} - \frac{1}{\theta} \gamma \psi$$

We solve this first-order differential equation and then use the boundary condition $\psi(\infty) = 0$ to find the following expression.

$$\psi(\theta) = \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_c(x)} f(x) - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx$$

We can isolate λ_1 using the boundary condition $\psi(0) = 0$.

$$\lambda_{1,t} = \frac{\int_0^{\infty} \exp \left[- \int_0^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_{c,t}(x)} f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx}{\int_0^{\infty} \exp \left[- \int_0^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \bar{\alpha}_t(x) f_t(x) dx}$$

Finally, substituting ψ and then λ_2 into the expression for the commodity wedge gives us an ABCD-style formula for the optimal commodity wedge.

$$\frac{\tau^{c_j}}{1 + \tau^{c_j}} = A_t(\theta)^{c_j} B_t(\theta) C_t(\theta) + \beta R \left(\frac{\tau_{t-1}^{c_j}}{1 + \tau_{t-1}^{c_j}} \right) D_t(\theta)^{c_j}$$

$$A_t(\theta)^{c_j} = \gamma_{1,t}(\theta) - \gamma_{j,t}(\theta)$$

$$B_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$C_t(\theta) = \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)}$$

$$D_t(\theta)^{c_j} = \frac{A_t(\theta)^{c_j}}{A_{t-1}^{c_j}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{1,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}$$

B Optimal wedge derivation with externalities

B.1 Commodity wedge

The Hamiltonian to this problem is as follows.

$$\begin{aligned}\mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta, S) \right] f \\ & + \psi \left[-U_l(c, q, y/\theta, S) \frac{l}{\theta} + \beta w_2 \right] \\ & - \lambda_1 u(\theta) f_t \\ & + \lambda_2 u(\theta) f_2 \\ & + \phi [u - U(c, q, y/\theta, S) - \beta w] \\ & - \mu [S - S(\dots, Q)]\end{aligned}$$

This yields the following first-order conditions.

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial c} : & f - \psi U_{cl} \frac{l}{\theta} = \phi U_c \\ \frac{\partial \mathcal{H}}{\partial q} : & f - \psi U_{ql} \frac{l}{\theta} + \mu S_q f = \phi U_q \\ \frac{\partial \mathcal{H}}{\partial l} : & -\theta f - \phi U_l = \frac{1}{\theta} \psi [U_{ll} l + U_l] \\ \frac{\partial \mathcal{H}}{\partial u} : & -\lambda_1 f + \lambda_2 f_2 + \phi = \dot{\psi} \\ \frac{\partial \mathcal{H}}{\partial w} : & \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f = \phi \beta \\ \frac{\partial \mathcal{H}}{\partial w_2} : & \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta \\ \frac{\partial \mathcal{H}}{\partial S} : & \int \left(-\psi U_{ls} \frac{l}{\theta} - \phi U_s + \frac{1}{R} \frac{\partial V_{t+1}}{\partial S} \right) f = \mu\end{aligned}$$

As in the previous proof,

$$\phi = \frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c}$$

Because every good j shares the same ϕ ,

$$\phi = \frac{f}{U_q} - \frac{\psi U_{ql} \frac{l}{\theta}}{U_q} + \frac{\mu S_q f}{U_q}$$

Hence,

$$\frac{f}{U_q} - \frac{\psi U_{ql} \frac{l}{\theta}}{U_q} + \frac{\mu S_q f}{U_q} = \frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c}$$

If we multiply by U_{c_j} and U_{c_1} ,

$$U_c \left(f - \psi U_{ql} \frac{l}{\theta} + \mu S_q f \right) = U_q \left(f - \psi U_{cl} \frac{l}{\theta} \right)$$

Using $\gamma_k \equiv \frac{U_{c_k l}}{U_{c_k}}$, or, equivalently, $\gamma_k U_{c_k} = U_{c_k l}$,

$$U_c \left(f - \frac{\psi}{\theta} \gamma_q U_q + \mu S_q f \right) = U_q \left(f - \frac{\psi}{\theta} \gamma_c U_c \right)$$

Then, putting the γ terms (as well as the externality term) on one side,

$$U_c f - U_q f = \frac{\psi}{\theta} U_c U_q (\gamma_q - \gamma_c) - \mu U_c S_q f$$

Finally, dividing by $U_q f$,

$$\frac{U_c}{U_q} - 1 = \frac{\psi U_c}{\theta f} (\gamma_q - \gamma_c) - \mu \frac{U_c}{U_q} S_q$$

Note that, by definition, $1 + \tau^q = U_q/U_c$, so $\frac{U_c}{U_q} - 1 = -\frac{\tau^q}{1 + \tau^q}$. Multiplying through by -1 ,

$$\frac{\tau^q}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q) + \mu \frac{U_c}{U_q} S_q$$

Since $\frac{U_c}{U_q} = \frac{1}{1+\tau^q}$,

$$\frac{\tau^q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q) + \frac{\mu S_q}{1+\tau^q}$$

If we rearrange terms,

$$\frac{\tau^q - \mu S_q}{1+\tau^q} = \frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$$

Hence, if we follow the same steps as appendix A.1,

$$\frac{\tau_t^q - \mu_t S_q}{1+\tau_t^q} = A_t(\theta)B_t(\theta)C_t(\theta) + \beta R \left(\frac{\tau_{t-1}^q - \mu_{t-1} S_{q,t-1}}{1+\tau_{t-1}^q} \right) D_t(\theta)$$

$$A_t(\theta) = \gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)$$

$$B_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$C_t(\theta) = \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)}$$

$$D_t(\theta) = \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{1,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}$$

Now let us focus on μS_q .

$$\int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} f - \psi U_{ls} \frac{l}{\theta} - \phi U_S \right) S_q$$

If we substitute ϕ .

$$\begin{aligned} \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} f - \psi U_{ls} \frac{l}{\theta} - \left[\frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c} \right] U_S \right) S_q &= \int \frac{U_c}{U_q} \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} f - \psi U_{ls} \frac{l}{\theta} - \left[f - \psi U_{cl} \frac{l}{\theta} \right] \frac{U_S}{U_c} \right) S_q \\ &= \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \frac{\psi}{\theta f} U_{ls} l - \left[1 - \frac{\psi}{\theta f} U_{cl} \right] \frac{U_S}{U_c} \right) S_q f \end{aligned}$$

We can rewrite $\psi/\theta f$ noting that

$$\frac{\psi}{\theta f} = \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{U_c} \frac{1}{1 + \epsilon - \gamma_c}$$

Hence, also defining $\xi \equiv \frac{U_{ls} l}{U_S}$,

$$\begin{aligned} \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \frac{\psi}{\theta f} U_{ls} l - \left[1 - \frac{\psi}{\theta f} U_{cl} \right] \frac{U_S}{U_c} \right) S_q f &= \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{1 + \epsilon - \gamma_c} \frac{U_{ls} l}{U_c} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{U_{cl}}{1 + \epsilon - \gamma_c} \right] \frac{U_S}{U_c} \right) S_q f \\ &= \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{1 + \epsilon - \gamma_c} \frac{U_{ls} l}{U_c} \frac{U_S}{U_S} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{U_{cl}}{1 + \epsilon - \gamma_c} \right] \frac{U_S}{U_c} \right) S_q f \\ &= \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\xi}{1 + \epsilon - \gamma_c} \frac{U_S}{U_c} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{U_{cl}}{1 + \epsilon - \gamma_c} \right] \frac{U_S}{U_c} \right) S_q f \\ &= \int \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \left[1 + \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\xi - \gamma_c}{1 + \epsilon - \gamma_c} \right] \frac{U_S}{U_c} \right) S_q f \end{aligned}$$

By the envelope theorem, we can iterate through $\frac{\partial V_{t+1}}{\partial S}$ to find that

$$\text{SCC}_t = - \int \left(\sum_{j=0}^s \left[1 + \left(\frac{\tau_j^y}{1 - \tau_j^y} \right) \frac{\xi_j - \gamma_{c,j}}{1 + \epsilon_j - \gamma_{c,j}} \right] \frac{U_{S,j}}{U_{c,j}} S_{s-j} \right) f$$

Here, s is the number of periods emission affect the pollution stock.

C Constrained-optimal wedge derivations

C.1 Constrained-optimal commodity wedge (suboptimal labor wedge)

$$V_t(\hat{w}, \hat{w}_2, \theta_-, Q_-) = \min_{c, q, y, u, w, w_2, S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, Q) \right] f_t(\theta | \theta_-) d\theta$$

s.t.

$$\dot{u}(\theta, S) = U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w_2(\theta)$$

$$\hat{w} = \int_0^\infty u(\theta) f_t(\theta | \theta_-) d\theta$$

$$\hat{w}_2 = \int_0^\infty u(\theta) f_{2,t}(\theta | \theta_-) d\theta$$

$$u(\theta, S) = U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w(\theta)$$

$$S = S \left(\dots, \int q(\theta) f(\theta) d\theta \right)$$

$$1 - \bar{\tau}^y(\theta) = - \frac{U_l \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right)}{\theta U_c \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right)}$$

This yields this Hamiltonian, where ν^y is the Lagrange multiplier on the labor wedge constraint.

$$\begin{aligned} \mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1} \left(w, w_2, \theta, \int q(\theta) f(\theta) d\theta \right) \right] f \\ & + \psi \left[-U_l(c, q, l, S) \frac{l}{\theta} + \beta w_2 \right] \\ & - \lambda_1 u(\theta) f_t \\ & + \lambda_2 u(\theta) f_2 \\ & + \phi [u - U(c, q, l, S) - \beta w] \\ & - \mu \left[S - S \left(\dots, \int q(\theta) f(\theta) d\theta \right) \right] \\ & + \nu^y \left[1 - \bar{\tau}^y(\theta) + \frac{U_l(c, q, l, S)}{\theta U_c(c, q, l, S)} \right] \end{aligned}$$

The first-order conditions are as follows.

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} + \nu^y \left(\frac{U_{cl}}{\theta U_c} - \frac{U_l U_{cc}}{\theta U_c^2} \right) &= \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f + \frac{1}{R} \frac{\partial V}{\partial Q} f(\theta) - \psi U_{ql} \frac{l}{\theta} + \mu S_2 f + \nu^y \left(\frac{U_{ql}}{\theta U_c} - \frac{U_l U_{cq}}{\theta U_c^2} \right) &= \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l + \nu^y \left(\frac{U_{ll}}{\theta U_c} - \frac{U_l U_{cl}}{\theta U_c^2} \right) &= \frac{1}{\theta} \psi [U_{ll} l + U_l] \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi &= \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f &= \phi \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f &= -\psi \beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int \left(-D'(S) \theta l f - \psi U_{ls} \frac{l}{\theta} - \phi U_S + \nu^y \left(\frac{U_{ls}}{\theta U_c} - \frac{U_l U_{cs}}{\theta U_c^2} \right) \right) - \frac{1}{R} \frac{\partial V_{t+1}}{\partial S} f = \mu
\end{aligned}$$

Using the labor wedge constraint $1 - \bar{\tau}^y(\theta) = -\frac{U_l}{\theta U_c}$,

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} - \nu^y (1 - \bar{\tau}^y(\theta)) \left(\frac{U_{cl}}{U_l} - \frac{U_{cc}}{U_c} \right) &= \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f + \frac{1}{R} \frac{\partial V}{\partial Q} f(\theta) - \psi U_{ql} \frac{l}{\theta} + \mu S_2 f - \nu^y (1 - \bar{\tau}^y(\theta)) \left(\frac{U_{ql}}{U_l} - \frac{U_{cq}}{U_c} \right) &= \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l - \nu^y (1 - \bar{\tau}^y(\theta)) \left(\frac{U_{ll}}{U_l} - \frac{U_{cl}}{U_c} \right) &= \frac{1}{\theta} \psi [U_{ll} l + U_l] \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi &= \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f &= \phi \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f &= -\psi \beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int \left(-D'(S) \theta l f - \psi U_{ls} \frac{l}{\theta} - \phi U_S - \nu^y (1 - \bar{\tau}^y(\theta)) \left(\frac{U_{ls}}{U_l} - \frac{U_{cs}}{U_c} \right) \right) - \frac{1}{R} \frac{\partial V_{t+1}}{\partial S} f = \mu
\end{aligned}$$

Defining the semi-elasticity of the labor wedge with respect to x ,

$$\begin{aligned}\eta_x &\equiv \frac{\partial}{\partial x} \log(1 - \bar{\tau}^y) = \frac{\partial}{\partial x} \log\left(-\frac{U_l}{\theta U_c}\right) \\ &= \frac{\partial}{\partial x} \log(-U_l) - \frac{\partial}{\partial x} \log(\theta U_c) \\ &= \frac{U_{lx}}{U_l} - \frac{U_{cx}}{U_c}\end{aligned}$$

Hence,

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} - \nu^y(1 - \bar{\tau}^y(\theta))\eta_c &= \phi U_c \\ \frac{\partial \mathcal{H}}{\partial q} &: f + \frac{1}{R} \frac{\partial V}{\partial Q} f(\theta) - \psi U_{ql} \frac{l}{\theta} + \mu S_2 f - \nu^y(1 - \bar{\tau}^y(\theta))\eta_q &= \phi U_q \\ \frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l - \nu^y(1 - \bar{\tau}^y(\theta))\eta_l &= \frac{1}{\theta} \psi [U_{ll} + U_l] \\ \frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi &= \dot{\psi} \\ \frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f &= \phi \beta \\ \frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f &= -\psi \beta \\ \frac{\partial \mathcal{H}}{\partial S} &: \int \left(-D'(S) \theta l f - \psi U_{ls} \frac{l}{\theta} - \phi U_s - \nu^y(1 - \bar{\tau}^y(\theta))\eta_s \right) - \frac{1}{R} \frac{\partial V_{t+1}}{\partial S} f &= \mu\end{aligned}$$

As in the earlier proofs, we note that both goods share the same ϕ .

$$\frac{f - \psi U_{cl} \frac{l}{\theta} - \nu^y(1 - \bar{\tau}^y(\theta))\eta_c}{U_c} = \frac{f + \frac{1}{R} \frac{\partial V}{\partial Q} f(\theta) - \psi U_{ql} \frac{l}{\theta} + \mu S_2 f - \nu^y(1 - \bar{\tau}^y(\theta))\eta_q}{U_q}$$

If we multiply by U_c and U_q ,

$$U_c \left(f + \frac{1}{R} \frac{\partial V}{\partial Q} f(\theta) - \psi U_{ql} \frac{l}{\theta} + \mu S_2 f - \nu^y(1 - \bar{\tau}^y(\theta))\eta_q \right) = U_q \left(f - \psi U_{cl} \frac{l}{\theta} - \nu^y(1 - \bar{\tau}^y(\theta))\eta_c \right)$$

In what follows, the $\frac{1}{R} \frac{\partial V}{\partial Q} f(\theta)$ piece from the FOC for q is absorbed into the modified social cost of carbon together with μS_q , so subsequent equations carry only the combined μS_q object.

Using $\gamma_k \equiv \frac{U_{c_k l}}{U_{c_k}}$, or, equivalently, $\gamma_k U_{c_k} = U_{c_k l}$,

$$U_c \left(f + \frac{1}{R} \frac{\partial V}{\partial Q} f(\theta) - \frac{\psi}{\theta} \gamma_q U_q + \mu S_2 f - \nu^y (1 - \bar{\tau}^y(\theta)) \eta_q \right) = U_q \left(f - \frac{\psi}{\theta} \gamma_c U_c - \nu^y (1 - \bar{\tau}^y(\theta)) \eta_c \right)$$

Then, putting the γ terms (as well as the externality and wedge constraint terms) on one side,

$$U_c f - U_q f = \frac{\psi}{\theta} U_c U_q (\gamma_q - \gamma_c) - \mu U_c S_q f + \nu^y (1 - \bar{\tau}^y(\theta)) (\eta_q U_c - \eta_c U_q)$$

Alternatively, using $U_q/U_c = 1 + \tau^q$,

$$U_c f - U_q f = \frac{\psi}{\theta} U_c U_q (\gamma_q - \gamma_c) - \mu U_c S_q f + \nu^y (1 - \bar{\tau}^y(\theta)) U_c (\eta_q - (1 + \tau^q) \eta_c)$$

Dividing by $U_q f$,

$$\begin{aligned} \frac{\tau^q}{1 + \tau^q} &= \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q) + \mu \frac{U_c}{U_q} S_q - \nu^y (1 - \bar{\tau}^y) \frac{U_c}{U_q f} (\eta_q - (1 + \tau^q) \eta_c) \\ &= \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q) + \frac{1}{1 + \tau^q} \left[\mu S_q - \frac{\nu^y}{f} (1 - \bar{\tau}^y) \eta_q \right] + \frac{\nu^y}{f} (1 - \bar{\tau}^y) \eta_c \end{aligned}$$

After rearranging,

$$\frac{\tau^q - \mu S_q}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q) + \frac{\nu^y}{f} (1 - \bar{\tau}^y) \eta_c - \frac{\nu^y}{f} \frac{1 - \bar{\tau}^y}{1 + \tau^q} \eta_q$$

Further,

$$\frac{\tau^q - \mu S_q + \frac{\nu^y}{f} (1 - \bar{\tau}^y) \eta_q}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q) + \frac{\nu^y}{f} (1 - \bar{\tau}^y) \eta_c$$

If we isolate ψ ,

$$\psi = \left[\frac{\tau^q - \mu S_q + \frac{\nu^y}{f}(1 - \bar{\tau}^y)\eta_q}{1 + \tau^q} - \frac{\nu^y}{f}(1 - \bar{\tau}^y)\eta_c \right] \frac{\theta f}{U_c(\gamma_c - \gamma_q)}$$

From here,

$$\lambda_{2,t+1} = \beta R \left[\frac{\tau_t^q - \mu S_{q,t} + \frac{\nu_t^y}{f_t}(1 - \bar{\tau}_t^y)\eta_{q,t}}{1 + \tau_t^q} - \frac{\nu_t^y}{f_t}(1 - \bar{\tau}_t^y)\eta_{c,t} \right] \frac{\theta_t}{U_{c,t}} \frac{1}{\gamma_{c,t} - \gamma_{q,t}}$$

To find the expression for $\psi(\theta)$, we can integrate $\frac{\partial \mathcal{H}}{\partial u}$ from θ to ∞ .

$$\begin{aligned} \dot{\psi} &= -\lambda_1 \bar{\alpha}_t f + \lambda_2 f_2 + \frac{f - \psi U_{cl} \frac{l}{\theta} - \nu^y(1 - \bar{\tau}^y(\theta))\eta_c}{U_c} \\ &= -\lambda_1 \bar{\alpha}_t f + \lambda_2 f_2 + \frac{f - \nu^y(1 - \bar{\tau}^y(\theta))\eta_c}{U_c} - \frac{1}{\theta} \psi \frac{U_{cl} l}{U_c} \\ &= -\lambda_1 \bar{\alpha}_t f + \lambda_2 f_2 + \frac{f - \nu^y(1 - \bar{\tau}^y(\theta))\eta_c}{U_c} - \frac{1}{\theta} \gamma \psi \end{aligned}$$

If we solve this first-order differential equation, we obtain the following.

$$\psi(\theta) = \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1 - \frac{\nu^y(x)}{f(x)}(1 - \bar{\tau}^y(x))\eta_c}{U_c(x)} f(x) - \lambda_1 f(x) + \lambda_2 f_2(x) \right) dx$$

Using the boundary condition $\psi(0) = 0$,

$$\lambda_{1,t} = \frac{\int_0^{\infty} \exp \left[- \int_0^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1 - \frac{\nu_t^y(x)}{f_t(x)}(1 - \bar{\tau}_t^y(x))\eta_{c,t}(x)}{U_{c,t}(x)} f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx}{\int_0^{\infty} \exp \left[- \int_0^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_t(x) dx}.$$

If we substitute ψ into the expression for the commodity wedge,

$$\frac{\tau^q - \mu S_q + \frac{\nu^y}{f}(1 - \bar{\tau}^y)\eta_q}{1 + \tau^q} = \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1 - \frac{\nu^y(x)}{f(x)}(1 - \bar{\tau}^y(x))\eta_c}{U_c(x)} f(x) - \lambda_1 f(x) + \lambda_2 f_2(x) \right) dx - \frac{U_c}{\theta f}(\gamma_c - \gamma_q) + \frac{\nu^y}{f}(1 - \bar{\tau}^y)\eta_c$$

We can rewrite this as following, after substituting λ_2 and rearranging terms.

$$\begin{aligned} \frac{\tau_t^q - \mu_t S_{q,t} + \frac{\nu_t^y}{f_t}(1 - \bar{\tau}_t^y)\eta_{q,t}}{1 + \tau_t^q} &= \int_{\theta}^{\infty} \frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(1 - \frac{\nu_t^y(x)}{f_t(x)}(1 - \bar{\tau}_t^y(x))\eta_{c,t}(x) \right) (\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)) \frac{f_t(x)}{\theta_t f_t(\theta)} dx - \\ &\quad \int_{\theta}^{\infty} \frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] U_{c,t}(x) (\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)) \lambda_1 \bar{\alpha}_t(x) \frac{f_t(x)}{\theta_t f_t(\theta)} dx + \\ &\quad \beta R \left(\frac{\tau_{t-1}^q - \mu_{t-1} S_{q,t-1} + \frac{\nu_{t-1}^y}{f_{t-1}}(1 - \bar{\tau}_{t-1}^y)\eta_{q,t-1}}{1 + \tau_{t-1}^q} - \frac{\nu_{t-1}^y}{f_{t-1}}(1 - \bar{\tau}_{t-1}^y)\eta_{c,t-1} \right) \\ &\quad \frac{\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)}{\gamma_{c,t-1}(\theta) - \gamma_{q,t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)} + \\ &\quad \frac{\nu_t^y}{f_t}(1 - \bar{\tau}_t^y)\eta_{c,t} \end{aligned}$$

Alternatively,

$$\begin{aligned} \frac{\tau_t^q - \mu_t S_{q,t} + \frac{\nu_t^y}{f_t}(1 - \bar{\tau}_t^y)\eta_{q,t}}{1 + \tau_t^q} &= \int_{\theta}^{\infty} \frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(1 - \frac{\nu_t^y(x)}{f_t(x)}(1 - \bar{\tau}_t^y(x))\eta_{c,t}(x) \right) (\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)) \frac{f_t(x)}{\theta_t f_t(\theta)} dx - \\ &\quad \int_{\theta}^{\infty} \frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] U_{c,t}(x) (\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)) \lambda_1 \bar{\alpha}_t(x) \frac{f_t(x)}{\theta_t f_t(\theta)} dx + \\ &\quad \beta R \left(\frac{\tau_{t-1}^q - \mu_{t-1} S_{q,t-1} + \frac{\nu_{t-1}^y}{f_{t-1}}(1 - \bar{\tau}_{t-1}^y)\eta_{q,t-1}}{1 + \tau_{t-1}^q} \right) \\ &\quad \frac{\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)}{\gamma_{c,t-1}(\theta) - \gamma_{q,t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)} + \\ &\quad \frac{\nu_t^y}{f_t}(1 - \bar{\tau}_t^y)\eta_{c,t} - \\ &\quad \beta R \frac{\nu_{t-1}^y}{f_{t-1}}(1 - \bar{\tau}_{t-1}^y)\eta_{c,t-1} \frac{\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)}{\gamma_{c,t-1}(\theta) - \gamma_{q,t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)} \end{aligned}$$

After rewriting and rearranging,

$$\begin{aligned}
\frac{\tau_t^q - \mu_t S_{q,t} + \frac{\nu_t^y}{f_t}(1 - \bar{\tau}_t^y)\eta_{q,t}}{1 + \tau_t^q} &= (\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)) \frac{1 - F_t(\theta)}{\theta_t f_t(\theta)} \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] \\
&\quad \left(1 - \frac{\nu_t^y(x)}{f_t(x)} (1 - \bar{\tau}_t^y(x)) \eta_{c,t}(x) - \lambda_{1,t} \bar{\alpha}_{1,t}(x) U_{c,t}(x) \right) \frac{f_t(x)}{1 - F_t(\theta)} dx + \\
&\quad \beta R \left(\frac{\tau_{t-1}^q - \mu_{t-1} S_{q,t-1} + \frac{\nu_{t-1}^y}{f_{t-1}}(1 - \bar{\tau}_{t-1}^y)\eta_{q,t-1}}{1 + \tau_{t-1}^q} \right) \\
&\quad \frac{\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)}{\gamma_{c,t-1}(\theta) - \gamma_{q,t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)} + \\
&\quad \frac{\nu_t^y}{f_t} (1 - \bar{\tau}_t^y) \eta_{c,t} - \\
&\quad \beta R \frac{\nu_{t-1}^y}{f_{t-1}} (1 - \bar{\tau}_{t-1}^y) \eta_{c,t-1} \frac{\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)}{\gamma_{c,t-1}(\theta) - \gamma_{q,t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)}
\end{aligned}$$

This gives us the following modified ABCD form.

$$\begin{aligned}
\frac{\tau^q - \mu S_q + \frac{\nu^y}{f}(1 - \bar{\tau}^y)\eta_q}{1 + \tau^q} &= A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \left(\frac{\tau_{t-1}^q - \mu S_{q,t-1} + \frac{\nu_{t-1}^y}{f_{t-1}}(1 - \bar{\tau}_{t-1}^y)\eta_{q,t-1}}{1 + \tau_{t-1}^q} \right) D_t(\theta) + \\
&\quad \frac{\nu_t^y}{f_t} (1 - \bar{\tau}_t^y) \eta_{c,t} - \beta R \frac{\nu_{t-1}^y}{f_{t-1}} (1 - \bar{\tau}_{t-1}^y) \eta_{c,t-1} D_t(\theta)
\end{aligned}$$

$$A_t(\theta) = \gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)$$

$$B_t(\theta) = \frac{1 - F_t(\theta)}{\theta f_t(\theta)}$$

$$C_t(\theta) = \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] \left(1 - \frac{\nu_t^y}{f_t(x)} (1 - \bar{\tau}_t^y(x)) \eta_{c,t}(x) - \lambda_{1,t} \bar{\alpha}_{1,t}(x) U_{c,t}(x) \right) \frac{f_t(x) dx}{1 - F_t(\theta)}$$

$$D_t(\theta) = \frac{A_t(\theta)}{A_{t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}$$

Compact form. The two trailing ν^y -pieces appended to the modified ABCD form above — the contemporaneous $+\frac{\nu_t^y}{f_t}(1 - \bar{\tau}_t^y)\eta_{c,t}$ and the lagged $-\beta R \frac{\nu_{t-1}^y}{f_{t-1}}(1 - \bar{\tau}_{t-1}^y)\eta_{c,t-1} D_t(\theta)$ — are the date- t and date- $(t-1)$ copies of the same object. Subtracting $\frac{\nu_t^y}{f_t}(1 - \bar{\tau}_t^y)\eta_{c,t}$ from both sides and combining over the common denominator $1 + \tau_t^q$ folds them into the left-hand side, leaving a recursion that is structurally identical to the unconstrained ABCD identity of appendix B.1.

Define the bundled labor-wedge object

$$\bar{\Gamma}_{q,t}^y \equiv \frac{1 - \bar{\tau}_t^y}{f_t} \left[\eta_{c,t} - \frac{\eta_{q,t}}{1 + \tau_t^q} \right],$$

and the constrained-effective commodity wedge

$$\Theta_t^q \equiv \frac{\tau_t^q - \mu_t S_{q,t}}{1 + \tau_t^q} - \nu_t^y \bar{\Gamma}_{q,t}^y.$$

The modified ABCD form then collapses to the compact recursion

$$\boxed{\Theta_t^q = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \Theta_{t-1}^q D_t(\theta)} \quad (1)$$

with A_t , B_t , C_t , D_t as defined above. The C_t integrand still carries the within-tail correction $-\frac{\nu_t^y(x)}{f_t(x)}(1 - \bar{\tau}_t^y(x))\eta_{c,t}(x)$; that piece operates at types $x > \theta$ and cannot be absorbed at the local type without redefining C_t .

The compact form in (1) shows that the constrained-optimal commodity wedge inherits the entire dynamic architecture of Result 5.0.1, with the labor-wedge constraint entering as two and only two additive corrections — one local, one non-local. The intratemporal recursion $A_t(\theta)B_t(\theta)C_t(\theta)$, the intertemporal persistence $\beta R D_t(\theta)$, the hazard-ratio cost-benefit ratio B_t , the cross-good complementarity differential $A_t = \gamma_c - \gamma_q$ — all are structurally unchanged. The constraint shows up exclusively through a shift in the effective-wedge object Θ_t^q on both sides of the recursion (the local correction) and through a single extra term inside the C_t integrand (the non-local correction). Because the recursion operates on Θ_t^q at both dates, the constraint's correction at $t-1$ is propagated forward to t through the same kernel D_t that propagates yesterday's redistributive incentives, so the planner cannot separate the dynamic-incentive role of past commodity wedges from her dynamic accommodation of past labor-wedge constraints — they travel together through the same persistence channel.

The local correction $-\nu_t^y \bar{\Gamma}_{q,t}^y$ replaces the deviation $(\tau^q - \text{SCC})/(1 + \tau^q)$ of Result 5.0.1 with the labor-wedge-corrected effective wedge $\Theta_t^q = (\tau_t^q - \mu_t S_{q,t})/(1 + \tau_t^q) - \nu_t^y \bar{\Gamma}_{q,t}^y$. The bundle $\bar{\Gamma}_q^y = \frac{1 - \bar{\tau}^y}{f} \left[\eta_c - \frac{\eta_q}{1 + \tau^q} \right]$ compares how the labor keep rate $-U_l/(\theta U_c)$ responds to consumption

of the clean good (η_c) versus to consumption of the dirty good at the relative price ($\eta_q/(1 + \tau^q)$). When the c -channel dominates — the natural case under separability and most standard preferences — $\bar{\Gamma}_q^y > 0$, and under-distorted labor ($\nu_t^y > 0$) pulls the effective commodity wedge below its raw Pigou–Mirrleesian level $(\tau^q - \mu S_q)/(1 + \tau^q)$. The planner is moderating the commodity tax to relieve pressure on the constrained labor margin: holding c relatively cheap keeps U_c relatively low, which mechanically loosens the labor keep rate in the direction her constrained labor wedge cannot. When the q -channel dominates — leisure-complementary dirty goods, or dirty goods that strongly enter the marginal utility of clean consumption — the sign flips and the planner pushes the commodity wedge above the Pigou–Mirrleesian level, using the commodity instrument as a synthetic labor wedge. This is the dynamic, type-dependent generalisation of Jacobs and de Mooij (2015)’s “virtual subsidy” channel: the commodity wedge does the labor wedge’s work, and the magnitude of substitution is set entirely by the local semi-elasticity differential $\eta_c - \eta_q/(1 + \tau^q)$.

The non-local correction inside C_t operates through a structurally distinct channel and across a different range of types. Where the local correction acts at the type θ whose wedge is being computed, the new piece $-(\nu^y(x)/f(x))(1 - \bar{\tau}^y(x))\eta_c(x)$ inside the C_t integrand subtracts from the redistributive valuation $1 - \lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x)$ at *every higher type* $x > \theta$ where the labor-wedge constraint binds. Recall that C_t measures the social value of resources extracted from higher types when the planner relaxes the incentive constraint at θ ; the new piece records that at types where the planner cannot reach her preferred labor distortion, the Mirrleesian commodity-mix lever has less work it can usefully do. The Mirrleesian/CCH motive of Result 4.0.1 is *blunted* along the upper tail in proportion to how strongly and how far up the constraint binds, with the same exponential propagation factor as in Results 4.0.1 and 5.0.1 deciding how much the blunting at distant types $x \gg \theta$ matters for the wedge at θ . This is genuinely a non-local effect: a constraint that binds only at the top of the skill distribution still shifts the optimal commodity wedge for middle types, because the redistributive integral integrates upward from θ to infinity.

Three implications follow. First, the compact form nests Result 5.0.1 exactly: when $\nu_t^y(\theta) \equiv 0$ for every type and every period, both the local correction $-\nu^y\bar{\Gamma}_q^y$ and the non-local integrand piece $-\nu^y(x)\eta_c(x)(1 - \bar{\tau}^y(x))/f(x)$ vanish, Θ_t^q collapses to $(\tau^q - \mu S_q)/(1 + \tau^q)$, and the formula reproduces Result 5.0.1 without modification. The constrained-optimal problem inherits the second-best problem as a smooth limit, with the labor-wedge constraint contributing only additive deviations that disappear when the constraint is slack. Second, the constraint’s two channels are structurally separable and quantitatively independent: the local channel responds to ν^y at the type θ being analysed, while the non-local channel responds to the entire upper-tail profile $\{\nu^y(x)\}_{x>\theta}$. A binding constraint at the bottom of the skill distribution shifts Θ^q at low types but barely touches C_t at high types; a binding constraint at the top reshapes C_t for every type below the bite even when $\nu^y(\theta) = 0$ locally. The planner therefore has structural reason to track the entire profile of constraint failures across types,

not just its local value — a feature absent from the static dynamic-Mirrleesian models of Mirrlees (1971) and Christiansen (1984), and from the unconstrained dynamic formulations of Golosov et al. (2016) and Barrage (2019). Third, the persistence structure $\beta R \Theta_{t-1}^q D_t$ propagates yesterday’s constrained-corrected effective wedge as a single object: the planner’s intertemporal-incentive role for past commodity wedges and her dynamic accommodation of past labor-wedge constraints are carried by the same kernel and cannot be disentangled at the optimum. A planner whose labor wedge was under-distorted yesterday must adjust the commodity wedge today — through the sign-dependent $\bar{\Gamma}_q^y$ channel — to fulfil the dynamic-incentive recursion, and this dynamic substitution survives even into periods where the contemporaneous labor wedge happens to be set optimally ($\nu_t^y = 0$ but $\nu_{t-1}^y \neq 0$). The constrained-optimal carbon tax therefore inherits the same Pigouvian/Mirrleesian decomposition as Result 5.0.1, but with a third, fully dynamic “substitution” role grafted onto the persistence channel.

This intermediate expression mirrors the unconstrained Section B.1 result up to two new ν^y -weighted terms generated by the labor-wedge constraint. To bring the wedge into the ABCD form of Section 5, we (i) isolate ψ from the FOC for l , (ii) integrate the FOC for u to obtain $\psi(\theta)$ in closed form, (iii) compute the modified social cost of carbon $\widetilde{\text{SCC}}_t$ from the FOC for S , and (iv) substitute and apply the Golosov–Kocherlakota–Tsyvinski transformation.

Isolating ψ from the FOC for l . Substitute ϕ from the FOC for c into the FOC for l . Using $-U_l/(\theta U_c) = 1 - \bar{\tau}^y$ and the identity (obtained by multiplying $U_{kl}/(\theta U_c)$ by U_l/U_l)

$$\frac{U_{kl}}{\theta U_c} + (1 - \bar{\tau}^y) \frac{U_{ck}}{U_c} = -(1 - \bar{\tau}^y) \eta_k \quad k \in \{c, l, S\},$$

the FOC-for- c residual collapses to $\phi U_c = f - \psi \gamma_c U_c / \theta - \nu^y (1 - \bar{\tau}^y) \eta_c$. Substituting into the FOC for l and using $U_{cl} l = \gamma_c U_c$, $U_{ll} l / U_l = \epsilon$:

$$-\bar{\tau}^y \theta f + \psi (1 - \bar{\tau}^y) U_c (1 + \epsilon - \gamma_c) = (1 - \bar{\tau}^y) \nu^y [(1 - \bar{\tau}^y) \theta \eta_c + \eta_l]$$

Dividing by $(1 - \bar{\tau}^y) \theta f$ and defining

$$\Gamma_l^y \equiv \frac{\eta_l}{\theta f} + \frac{(1 - \bar{\tau}^y) \eta_c}{f},$$

we obtain

$$\frac{\bar{\tau}^y}{1 - \bar{\tau}^y} = \frac{\psi U_c}{\theta f} (1 + \epsilon - \gamma_c) - \nu^y \Gamma_l^y \iff \psi = \left[\frac{\bar{\tau}^y}{1 - \bar{\tau}^y} + \nu^y \Gamma_l^y \right] \frac{\theta f}{U_c (1 + \epsilon - \gamma_c)}.$$

Because $\psi = -\frac{1}{\beta R} \frac{\partial V_{t+1}}{\partial w_2} f$, the envelope condition $\lambda_{2,t} = -\partial V_t / \partial \hat{w}_2$ at the previous-period optimum yields the lagged multiplier

$$\lambda_{2,t} = \beta R \left[\frac{\bar{\tau}_{t-1}^y}{1 - \bar{\tau}_{t-1}^y} + \nu_{t-1}^y \Gamma_{l,t-1}^y \right] \frac{\theta_{t-1}}{U_{c,t-1}(\theta)} \frac{1}{1 + \epsilon_{t-1} - \gamma_{c,t-1}}.$$

Solving the ψ ODE. Substituting ϕ into the FOC for u ($\dot{\psi} = -\lambda_1 f + \lambda_2 f_2 + \phi$):

$$\underbrace{\frac{d\psi}{dx}}_{\frac{dy}{dx}} = \underbrace{-\lambda_1 f + \lambda_2 f_2 + \frac{f - \nu^y(1 - \bar{\tau}^y)\eta_c}{U_c}}_Q - \underbrace{\frac{\gamma_c}{\theta}}_{\frac{\theta}{P}} \underbrace{\psi}_y.$$

This is a first-order linear ODE. With integrating factor $\exp\left[\int \gamma_c(\tilde{x}) d\tilde{x}/\tilde{x}\right]$, the solution that satisfies the boundary condition at infinity is

$$\psi(\theta) = \int_{\theta}^{\infty} \exp\left[-\int_{\theta}^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}}\right] \left(\frac{1 - \nu^y(x) \Gamma_c^y(x)}{U_c(x)} f(x) - \lambda_1 f(x) + \lambda_2 f_2(x) \right) dx,$$

where I define

$$\Gamma_c^y(x) \equiv \frac{(1 - \bar{\tau}^y(x)) \eta_c(x)}{f(x)}.$$

Using the boundary condition $\psi(0) = 0$, we pin down $\lambda_{1,t}$:

$$\lambda_{1,t} = \frac{\int_0^{\infty} \exp\left[-\int_0^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}}\right] \left(\frac{1 - \nu_t^y(x) \Gamma_{c,t}^y(x)}{U_{c,t}(x)} f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx}{\int_0^{\infty} \exp\left[-\int_0^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}}\right] f_t(x) dx}.$$

Modified social cost of carbon. The FOC for S at this appendix's first set of FOCs picks up an additional $\nu^y[U_{lS}/(\theta U_c) + (1 - \bar{\tau}^y)U_{cS}/U_c] = -\nu^y(1 - \bar{\tau}^y)\eta_S$ piece by the same identity used above. Carrying this through the $\mu U_c S_q/U_q$ computation in exact analogy with appendix B.1 (and using $\psi U_c/(\theta f) = [\bar{\tau}^y/(1 - \bar{\tau}^y) + \nu^y \Gamma_l^y]/(1 + \epsilon - \gamma_c)$ from above to substitute out ψ in the $U_{lS} l$ term) yields

$$\widetilde{\text{SCC}}_t = \left(\frac{1}{R} \frac{\partial V_{t+1}}{\partial S} - \left[1 + \left(\frac{\bar{\tau}^y}{1 - \bar{\tau}^y} + \nu^y \Gamma_l^y \right) \frac{\xi - \gamma_c}{1 + \epsilon - \gamma_c} + \nu^y \tilde{\Gamma}_S^y \right] \frac{U_S}{U_c} \right) S_q f_t,$$

where $\xi \equiv U_{lS} l/U_S$ and

$$\tilde{\Gamma}_S^y \equiv \frac{(1 - \bar{\tau}^y) \eta_S}{f}.$$

The marginal-damages piece $D'(S) \theta l f_t$ from the FOC for S remains outside the ν^y correction — it sits in $\partial V_{t+1}/\partial S$ exactly as in the unconstrained Section 5 formula — and is collected inside the first term of $\widetilde{\text{SCC}}_t$.

Assembling the ABCD form. Substitute ψ and $\lambda_{2,t}$ into the corrected starting expression. Following the same Golosov–Kocherlakota–Tsyvinski (2003) transformation as in appendix B.1 that rewrites the $\exp[-\int \gamma_c d\tilde{x}/\tilde{x}] \cdot 1/U_c(x)$ kernel as $(1-F)/(\theta f) \cdot 1/U_c(\theta)$ times the $\exp[\int(\sigma\dot{c}/c - \gamma\dot{y}/y) d\tilde{x}]$ kernel, the wedge expression becomes:

$$\begin{aligned}
\frac{\tau_t^q - \widetilde{\text{SCC}}_t}{1 + \tau_t^q} &= \int_{\theta}^{\infty} \exp\left[-\int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}}\right] \left(\frac{1 - \nu_t^y(x) \Gamma_{c,t}^y(x)}{U_{c,t}(x)} f_t(x) - \lambda_{1,t} f_t(x) + \lambda_{2,t} f_{2,t}(x)\right) dx \cdot \frac{(\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)) U_{c,t}(\theta)}{\theta_t f_t(\theta)} + \nu_t^y \\
&= \dots \\
&= (\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)) \frac{1 - F_t(\theta)}{\theta_t f_t(\theta)} \int_{\theta}^{\infty} \exp\left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t}{c_t} - \gamma_t(\tilde{x}) \frac{\dot{y}_t}{y_t}\right) d\tilde{x}\right] (1 - \nu_t^y(x) \Gamma_{c,t}^y(x) - \lambda_{1,t} U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
&\quad + \beta R \left[\frac{\bar{\tau}_{t-1}^y}{1 - \bar{\tau}_{t-1}^y} + \nu_{t-1}^y \Gamma_{l,t-1}^y \right] \frac{\gamma_{c,t}(\theta) - \gamma_{q,t}(\theta)}{1 + \epsilon_{t-1} - \gamma_{c,t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp\left[-\int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x) dx}{\theta_t f_t(\theta)} \\
&\quad + \nu_t^y \Gamma_{q,t}^y
\end{aligned}$$

Result in ABCD form. Collecting terms,

$$\boxed{\frac{\tau_t^q - \widetilde{\text{SCC}}_t}{1 + \tau_t^q} = A_t^q(\theta) B_t(\theta) C_t^*(\theta) + \beta R \left[\frac{\bar{\tau}_{t-1}^y}{1 - \bar{\tau}_{t-1}^y} + \nu_{t-1}^y \Gamma_{l,t-1}^y \right] D_t^*(\theta) + \nu_t^y \Gamma_{q,t}^y} \quad (2)$$

Equivalent form with τ^q on both ends. The labor-wedge representation of the lagged bracket above is one of two equivalent ways to extract $\psi_{t-1} U_{c,t-1}/(\theta_{t-1} f_{t-1})$ from the period- $(t-1)$ FOCs. Using the commodity FOC instead — solving the corrected starting equation in this appendix for $\frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q)$ — gives the identity

$$\frac{\psi U_c}{\theta f}(\gamma_c - \gamma_q) = \frac{\tau^q - \widetilde{\text{SCC}}}{1 + \tau^q} - \nu^y \bar{\Gamma}_q^y, \quad \bar{\Gamma}_q^y \equiv \frac{1 - \bar{\tau}^y}{f} \left[\eta_c - \frac{\eta_q}{1 + \tau^q} \right], \quad (3)$$

where $\bar{\Gamma}_q^y$ is the *bundled* object used in the main-text proposition at section 6.2. Applying (3) at $t-1$ replaces the labor-wedge bracket of equation (2) with its commodity-wedge equivalent; the factor $1/(\gamma_c - \gamma_q)|_{t-1} = 1/A_{t-1}^q$ then combines with A_t^q in the persistence numerator to recover Section 5's D_t^q , giving the symmetric form

$$\boxed{\frac{\tau_t^q - \widetilde{\text{SCC}}_t}{1 + \tau_t^q} = A_t^q(\theta) B_t(\theta) C_t^*(\theta) + \beta R \left[\frac{\tau_{t-1}^q - \widetilde{\text{SCC}}_{t-1}}{1 + \tau_{t-1}^q} - \nu_{t-1}^y \bar{\Gamma}_{q,t-1}^y \right] D_t^q(\theta) + \nu_t^y \Gamma_{q,t}^y} \quad (4)$$

with

$$D_t^q(\theta) \equiv \frac{A_t^q(\theta)}{A_{t-1}^q(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)}$$

matching the unconstrained D_t^q from Section 5 exactly. Both boxed equations describe the same constrained optimum: equation (2) makes the *shadow labor wedge* interpretation transparent (Force 2 is yesterday's wish-list labor wedge propagated forward), while equation (4) gives visual parity with Section 5 and is the natural object when comparing across the dynamic-Mirrlees commodity-tax literature.

Notational caveat. The lagged bracket in equation (4) uses the bundled $\bar{\Gamma}_q^y$ (which packages the η_c piece together with the η_q piece at the local lagged type), while the contemporaneous additive term still uses the unbundled Γ_q^y (only the η_q piece, with the η_c contribution living *inside* C_t^* 's integrand across the upper tail of types $x > \theta$). This asymmetry is unavoidable: η_c at types $x \neq \theta$ structurally sits inside the integral that defines C_t^* and cannot be re-bundled at the local type without changing what C_t^* means.

where

$$\begin{aligned}
A_t^q(\theta) &\equiv \gamma_{c,t}(\theta) - \gamma_{q,t}(\theta) \\
B_t(\theta) &\equiv \frac{1 - F_t(\theta)}{\theta_t f_t(\theta)} \\
C_t^*(\theta) &\equiv \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \nu_t^y(x) \Gamma_{c,t}^y(x) - \lambda_{1,t} U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
D_t^*(\theta) &\equiv \frac{A_t^q(\theta)}{1 + \epsilon_{t-1} - \gamma_{c,t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)}
\end{aligned}$$

and the constraint-driven Γ objects collect the multiplier-weighted semi-elasticities of the labor wedge $\eta_x \equiv \partial_x \log(-U_l/(\theta U_c)) = U_{lx}/U_l - U_{cx}/U_c$:

$$\begin{aligned}
\Gamma_{l,t}^y &\equiv \frac{\eta_{l,t}}{\theta_t f_t} + \frac{(1 - \bar{\tau}_t^y) \eta_{c,t}}{f_t} \\
\Gamma_{c,t}^y(x) &\equiv \frac{(1 - \bar{\tau}_t^y(x)) \eta_{c,t}(x)}{f_t(x)} \\
\Gamma_{q,t}^y &\equiv - \frac{(1 - \bar{\tau}_t^y) \eta_{q,t}}{(1 + \tau_t^q) f_t} \\
\bar{\Gamma}_{q,t}^y &\equiv \frac{1 - \bar{\tau}_t^y}{f_t} \left[\eta_{c,t} - \frac{\eta_{q,t}}{1 + \tau_t^q} \right] \quad (\text{bundled; matches the main-text proposition's } \Gamma_q^y)
\end{aligned}$$

When $\nu_t^y(\theta) \equiv 0$ (i.e. the labor wedge happens to be set at its second-best level for every type and every period), $\Gamma_{l,t}^y$, $\Gamma_{c,t}^y$, $\Gamma_{q,t}^y$ and the ν^y piece of $\widehat{\text{SCC}}_t$ all vanish, and equation (2) collapses exactly to the unconstrained Section 5 result.

Interpretation. The left-hand side of equation (2) is the *effective* (carbon) commodity wedge: the actual wedge τ_t^q on the polluting good, net of the modified social cost of carbon $\widetilde{\text{SCC}}_t$, normalized by $(1 + \tau_t^q)$. Compared to the second-best SCC_t of Section 5, $\widetilde{\text{SCC}}_t$ replaces the labor-wedge multiplier $\tau^y/(1 - \tau^y)$ with the *shadow* labor wedge $\bar{\tau}^y/(1 - \bar{\tau}^y) + \nu^y \Gamma_l^y$ and adds a $\nu^y \bar{\Gamma}_S^y$ piece capturing how pollution shifts the keep rate of consumption that the constrained labor wedge cannot reach.

The right-hand side decomposes the effective wedge into three forces:

1. **Mirrleesian/Corlett–Hague–Christiansen base** $A_t^q B_t C_t^*$. The classical relative-complementarity term $\gamma_c - \gamma_q$ multiplied by the inverse Mirrleesian hazard $(1 - F_t)/(\theta f_t)$, integrated over higher types. The integrand of C_t^* now carries an extra $-\nu^y(x) \Gamma_c^y(x)$ piece: across all types $x > \theta$ the planner accounts for how the labor-wedge constraint at x reshapes the keep rate of clean consumption that would otherwise be reached through the labor wedge.
2. **Persistence through the shadow labor wedge** $\beta R[\bar{\tau}_{t-1}^y/(1 - \bar{\tau}_{t-1}^y) + \nu_{t-1}^y \Gamma_{l,t-1}^y] D_t^*$. The lagged object is the previous period's *constrained* labor wedge plus its ν^y -weighted correction. The persistence multiplier D_t^* replaces the second-best A_{t-1}^q denominator with $1 + \epsilon_{t-1} - \gamma_{c,t-1}$ because the lagged term is now expressed in labor-wedge format rather than commodity-wedge format. This force is alive whenever skills are persistent ($f_{2,t} \neq 0$); under i.i.d. skills it vanishes.
3. **Direct composition rescue** $\nu_t^y \Gamma_{q,t}^y$. A purely contemporaneous, type-by-type response: when the labor wedge is too low for type θ ($\nu^y > 0$), the planner can substitute by adjusting the relative price of q to indirectly tax labor through consumption-composition channels. Because $\Gamma_{q,t}^y$ depends only on $\eta_{q,t}$, this force survives even when utility is fully separable in c and l (where the CCH base in Force 1 vanishes), making it the cleanest empirical lever.

Cross-check with the main-text proposition. The main-text proposition in section 6.2 writes the result in an integrated Δ_t form rather than the recursive ABCD form derived here. The two are equivalent. Starting from the symmetric form (4), iterate the persistence term backward in time — each iteration brings in another $(\tau_{t-2}^q - \widetilde{\text{SCC}}_{t-2})/(1 + \tau_{t-2}^q) - \nu_{t-2}^y \bar{\Gamma}_{q,t-2}^y$ via the same identity (3) — then re-bundle the $\nu^y \Gamma_c^y$ piece of C_t^* together with $\nu^y \Gamma_q^y$ into the same single object $\bar{\Gamma}_q^y$ used in the proposition. Substituting the $\widetilde{\text{SCC}}_t$ expansion into the LHS and collecting the resulting time-summation produces Δ_t exactly. The bundled $\bar{\Gamma}_q^y$ in the lagged bracket of equation (4) is therefore not an arbitrary notational choice — it is precisely the object the proposition needs once the recursion is unrolled.

Reading the Γ terms. Each Γ object packages how a specific semi-elasticity of the keep rate $1 - \bar{\tau}^y = -U_l/(\theta U_c)$ enters the constrained-optimal commodity wedge. Recall $\eta_x \equiv \partial_x \log(-U_l/(\theta U_c)) = U_{lx}/U_l - U_{cx}/U_c$, with the standard convention $U_l < 0$, $U_c > 0$, $U_{ll} < 0$ (concavity in labor) so that the inverse Frisch elasticity $\epsilon = U_{ll}l/U_l > 0$. As an anchoring benchmark, take fully separable preferences $U(c, q, l, S) = u(c) + v(q) + h(l) + g(S)$, which kill all cross-derivatives between distinct arguments: $\gamma_c = \gamma_q = \xi = 0$, $\eta_q = \eta_S = 0$, $\eta_l = U_{ll}/U_l > 0$ and $\eta_c = -U_{cc}/U_c > 0$.

Γ_l^y — **the shadow labor wedge correction.** The bracket $[\bar{\tau}^y/(1 - \bar{\tau}^y) + \nu^y \Gamma_l^y]$ is the labor wedge the planner would set were the constraint relaxed: it is the imposed wedge plus the multiplier on the constraint, expressed in labor-wedge units. In the separable benchmark both pieces of Γ_l^y are positive — raising labor or raising consumption each pushes the keep rate up, holding the other fixed — so $\Gamma_l^y > 0$ unambiguously, and a positive multiplier ($\nu^y > 0$, the constrained labor wedge is too low) means the shadow wedge *exceeds* the imposed one. Force 2 then *raises* the commodity wedge in the next period: the lagged-shadow-wedge bracket is the input to the persistence multiplier D_{t+1}^* , so under-distorting labor today forces the planner to over-distort goods tomorrow. The same bracket sits inside $\widetilde{SCC}_t$, scaling the equity-adjusted Pigouvian piece: under-distorted labor today also means the SCC the planner targets is closer to the unscaled Pigouvian level (rather than the more aggressive equity-adjusted one). Under non-separable preferences η_c also picks up U_{lc}/U_l : if clean consumption is strongly Edgeworth-complementary with labor ($U_{lc} > 0$), the shadow wedge correction is dampened, and in extreme cases Γ_l^y can flip sign, allowing the inequality $\nu^y > 0$ to actually *lower* the commodity wedge in the persistence channel.

$\Gamma_c^y(x)$ — **the within-integrand absorption.** This is the only Γ term that varies along the type distribution. It enters the integrand of C_t^* as $-\nu^y(x)\Gamma_c^y(x)$, so the question is what happens to $|C_t^*|$ when the labor-wedge constraint binds at types $x > \theta$. In the separable benchmark $\Gamma_c^y(x) > 0$ for every x , and a positive $\nu^y(x)$ *shrinks* the integrand and hence $|C_t^*|$. The Mirrleesian/CCH motive for a positive commodity wedge on the polluting good ($\gamma_c > \gamma_q$) operates by mimicking labor-wedge progressivity through composition; when the labor wedge is constrained too low at higher types, that mimicking is partially pre-empted because the binding constraint has already absorbed some of the shadow value at those types. The planner therefore pulls back on Force 1 in proportion to how strongly the constraint binds across higher types — a *decrease* in the magnitude of the commodity wedge contribution from this channel, with the size of the pullback an integral over the entire upper tail rather than a local object at θ . With non-separable preferences the same pullback applies whenever $\eta_c(x) > 0$. If clean consumption is so strongly Edgeworth-complementary with labor that $\eta_c(x) < 0$, the sign flips and the constraint actually *amplifies* the Mirrleesian channel: under-distorted labor at higher types makes the planner push the commodity wedge *more*, not less, through Force 1.

Γ_q^y — **the direct composition rescue.** In the separable benchmark $\eta_q = 0$, so $\Gamma_q^y = 0$ in this decomposition and Force 3 is silent. (This is a regrouping choice: the main-text proposition bundles the η_c piece of Force 1 into a single Γ_q^y that does not vanish under separability.) The total wedge is the same; what differs is which channel carries the separable-case correction.) The substantive case is non-separable preferences. The sign of $\eta_q = U_{lq}/U_l - U_{cq}/U_c$ is the key object: it tells us whether more polluting-good consumption, holding c and l fixed, raises ($\eta_q > 0$) or lowers ($\eta_q < 0$) the keep rate at the margin. Suppose the dirty good is leisure-complementary: $U_{lq} > 0$ (extra q makes the marginal disutility of labor smaller in absolute value). With $U_l < 0$ this gives $U_{lq}/U_l < 0$ and, if cross-effects with c are weak, $\eta_q < 0$, hence $\Gamma_q^y > 0$. When $\nu^y > 0$ (labor under-distorted), the planner taxes the dirty good *more* than the second-best level. The mechanism is direct: by suppressing q the planner mechanically raises the keep rate, which substitutes for the labor wedge they cannot impose. The opposite case (q consumption-complementary, $U_{cq} > 0$, $\eta_q > 0$, $\Gamma_q^y < 0$) flips the sign: now *taxing* q would push the keep rate further down, reinforcing the under-distortion of labor; the planner instead *undertaxes* the dirty good below the second-best level, accepting more pollution to relax the labor margin. Force 3 is purely contemporaneous and type-by-type — it depends on η_q at θ , not at the upper tail — which is what makes it the cleanest empirical lever.

$\tilde{\Gamma}_S^y$ — **the pollution-stock SCC modifier.** This sits inside the bracket multiplying U_S/U_c in $\widehat{SCC}_t$, so it modifies the planner's virtual subsidy on environmental quality. In the separable benchmark $\eta_S = 0$ and $\tilde{\Gamma}_S^y = 0$, so $\widehat{SCC}_t$ reduces to the standard expression with the equity-adjustment scalar simply replaced by the shadow labor wedge from Γ_l^y . Under non-separable preferences, $\eta_S = U_{lS}/U_l - U_{cS}/U_c$ measures whether higher pollution raises or lowers the keep rate. A natural case: pollution makes labor more painful at the margin ($U_{lS} < 0$, with $U_l < 0$ giving $U_{lS}/U_l > 0$) while leaving marginal utility of consumption roughly unchanged, so $\eta_S > 0$ and $\tilde{\Gamma}_S^y > 0$. When $\nu^y > 0$, the planner softens the corrective penalty on q that flows from the utility-damage channel: the binding labor-wedge constraint already over-discourages labor on the high-pollution margin, so adding more environmental correction would compound the distortion. The opposite happens when pollution sharply depresses the marginal utility of consumption ($U_{cS} \ll 0$): η_S flips negative, $\tilde{\Gamma}_S^y < 0$, and the planner *strengthens* the virtual subsidy — under-distorted labor and pollution-suppressed marginal utility now reinforce each other, demanding a more aggressive corrective wedge.

Putting it together. The four Γ objects partition the constrained-optimal response into a clean economic decomposition. Γ_l^y is the planner's wish-list labor wedge, propagated through time (Force 2) and through the SCC. Γ_c^y governs how that wish list erodes the Mirrleesian/CCH channel along the upper tail of types (Force 1). Γ_q^y is the only intratemporal lever the planner has left to substitute directly for the constrained labor wedge (Force 3). And $\tilde{\Gamma}_S^y$ is the pollution-specific extension of the same logic to the environmental-correction

motive (inside the SCC). In the separable benchmark only Γ_l^y and Γ_c^y are active, and both have unambiguous positive signs — providing a clean, calibratable departure from the second-best: under-distorted labor today simultaneously *raises* the persistence contribution and *lowers* the Mirrleesian contribution, with the net direction depending on the relative strength of the two channels and on the sign of $A^q = \gamma_c - \gamma_q$. Under non-separable preferences, Γ_q^y and $\tilde{\Gamma}_S^y$ switch on, with signs hinging on the cross-derivatives $U_{lq}, U_{cq}, U_{lS}, U_{cS}$ that researchers would need to estimate to take the formula to the data.

When the Γ terms flip sign. Each Γ object inherits its sign from a specific Edgeworth cross-derivative — U_{cl} for Γ_l^y and Γ_c^y , U_{lq} vs U_{cq} for Γ_q^y , and U_{ls} vs U_{cs} for $\tilde{\Gamma}_s^y$. A “sign flip” occurs when a cross-derivative becomes large enough (in absolute value, with the appropriate sign) to overturn the contribution of the diagonal Hessian terms in η_x . When that happens, the corresponding force *reverses* the direction in which the labor-wedge constraint is transmitted into the commodity wedge: a channel that lowered the wedge under separability now raises it (or vice versa). Below I walk through each regime that flips a Γ and what the flipped sign means.

Strong U_{cl} and the Γ_l^y, Γ_c^y flips. The cross-derivative U_{cl} measures Edgeworth complementarity between clean consumption and labor: $U_{cl} > 0$ means the marginal utility of consumption rises with labor (work-related goods, commuting expenses, take-out food after long shifts) and equivalently that the marginal disutility of labor falls with consumption. As U_{cl} rises from zero, $U_{cl}/U_c > 0$ subtracts from $\eta_l = U_{ll}/U_l - U_{cl}/U_c$, while $U_{lc}/U_l < 0$ (since $U_l < 0$) subtracts from $\eta_c = U_{lc}/U_l - U_{cc}/U_c$. Both η_l and η_c shrink; past a threshold, both turn negative. Geometrically, in the strongly complementary regime working less also lowers the marginal utility of consumption, so the keep rate $-U_l/(\theta U_c)$ moves in the direction opposite to its separable-benchmark response: less labor now *raises* the keep rate (numerator falls more slowly than denominator does), and less consumption now *raises* it too.

What this implies for the wedge: with $\eta_c < 0$ and $\nu^y(x) > 0$ at higher types, the term $-\nu^y(x)\Gamma_c^y(x)$ inside the integrand of C_t^* becomes *positive*, so $|C_t^*|$ *grows* rather than shrinks. Force 1 amplifies: when consumption strongly complements labor and the labor wedge is constrained too low at higher types, the planner uses the commodity wedge *more aggressively* as a substitute labor-suppression instrument (suppressing c via the wedge mechanically suppresses labor through the complementarity). And with $\Gamma_l^y < 0$ and $\nu^y > 0$, the bracket $\bar{\tau}^y/(1 - \bar{\tau}^y) + \nu^y\Gamma_l^y$ becomes *smaller* than the imposed wedge alone: Force 2 transmits less labor-wedge content into the next period’s commodity wedge, not more, and the equity adjustment inside $\widehat{SCC}_t$ is dampened. Together, the two flips imply a regime where strong work–consumption complementarity makes the constrained problem behave as if the labor wedge were too *high*: Mirrleesian channel up, persistence channel down, even though $\nu^y > 0$ would naively read as “too low.” The sign of ν^y remains the multiplier on the constraint, but the channels through which it transmits have inverted, so the comparative statics one would write down from the separable benchmark are unreliable in this regime.

Strong U_{lq} vs U_{cq} and the Γ_q^y flip. Force 3 is silent under separability and has no natural sign — it is governed entirely by $\eta_q = U_{lq}/U_l - U_{cq}/U_c$. Two cases dominate.

$U_{lq} > 0$ *dominates* — “leisure-complementary q .” The polluting good makes labor at the margin less painful: think of transportation that supports commuting, a take-away meal that frees up time, or work-related amenities. Then $U_{lq}/U_l < 0$, $\eta_q < 0$, and $\Gamma_q^y > 0$. With $\nu^y > 0$, Force 3

says: tax the dirty good *above* second-best. Suppressing q raises the marginal disutility of labor, lowers $-U_l$, and thereby raises the keep rate — mechanically substituting for the labor wedge the planner cannot impose.

$U_{cq} > 0$ *dominates* — “consumption-complementary q .” The polluting good is part of a wider consumption bundle: gasoline with cars, electricity with appliances, packaged goods with refrigeration. Then $U_{cq}/U_c > 0$, $\eta_q > 0$, $\Gamma_q^y < 0$. With $\nu^y > 0$, Force 3 reverses: the planner taxes q *below* second-best. Suppressing q would now raise U_c further, lowering the keep rate, and *reinforcing* the under-distortion of labor; the planner accepts more pollution to relax the labor margin.

The flip is therefore between leisure-complementary q (planner over-taxes when $\nu^y > 0$) and consumption-complementary q (planner under-taxes). Empirically a researcher cannot infer the sign of Force 3 from labor-wedge data alone — they need to identify whether the polluting good substitutes for or complements work-related disutility. For most carbon-emitting goods (electricity, gasoline, heating fuel) the consumption-complementary case is arguably the empirically relevant one, suggesting Force 3 typically operates in the “under-tax” direction when the labor wedge is constrained too low.

Strong U_{lS} vs U_{cS} and the $\tilde{\Gamma}_S^y$ flip. The same logic applies to $\tilde{\Gamma}_S^y$, with pollution stock S in place of q and $\eta_S = U_{lS}/U_l - U_{cS}/U_c$.

$U_{lS} < 0$ *dominates* — *pollution makes labor more painful at the margin*. $U_{lS}/U_l > 0$, $\eta_S > 0$, $\tilde{\Gamma}_S^y > 0$. With $\nu^y > 0$ inside $\widehat{SCC}_t$, the planner softens the corrective penalty: pollution already over-discourages labor, and adding more environmental correction would compound the distortion when the labor wedge cannot be raised to compensate.

$U_{cS} \ll 0$ *dominates* — *pollution sharply suppresses the marginal utility of consumption*. η_S flips negative, $\tilde{\Gamma}_S^y < 0$. With $\nu^y > 0$, the planner *strengthens* the virtual subsidy. The pollution-induced collapse of consumption value and the constrained-too-low labor wedge reinforce each other, and a more aggressive corrective wedge is the only remaining way to restore the equity-adjusted optimum.

Synthesis of the sign-flip analysis. Each sign flip is driven by one specific cross-derivative becoming large enough (relative to the diagonal Hessian) to overturn the separable-benchmark sign of an η_x object. The flip then reverses the comparative statics of the corresponding force: a channel that lowered the commodity wedge now raises it, or vice versa. Crucially, the four channels are governed by *different* cross-derivatives (U_{cl} , U_{lq} , U_{cq} , U_{lS} , U_{cS}), so they can flip independently. A calibration in which $U_{cl} \approx 0$ (the standard textbook starting point) but U_{cq} is meaningfully positive — plausible for most carbon-emitting goods — will keep Γ_l^y and Γ_c^y in their separable-benchmark direction (Force 2 up, Force 1 down) while flipping Γ_q^y to the

under-tax direction. Empirically, taking the constrained-optimal commodity wedge to data therefore requires not just labor- and consumption-side elasticities, but identification of every Edgeworth cross-derivative that links the polluting good and pollution stock to consumption and labor at the margin.

C.2 Constrained-optimal commodity wedge (suboptimal savings wedge)

$$\begin{aligned}
V_t(\hat{w}, \hat{w}_2, \theta_-, Q_-) &= \min_{c, q, y, u, w, w_2, S} \int_0^\infty \left[c(\theta) + q(\theta) - D(S)y(\theta) + \frac{1}{R}V_{t+1}(w(\theta), w_2(\theta), \theta, Q) \right] f_t(\theta|\theta_-) d\theta \\
&\quad \text{s.t.} \\
\dot{u}(\theta, S) &= U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{D(S)\theta}, S \right) + \beta w_2(\theta) \\
\hat{w} &= \int_0^\infty u(\theta) f_t(\theta|\theta_-) d\theta \\
\hat{w}_2 &= \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-) d\theta \\
u(\theta, S) &= U \left(c(\theta), q(\theta), \frac{y(\theta)}{D(S)\theta}, S \right) + \beta w(\theta) \\
S &= S \left(\dots, \int q(\theta) f(\theta) d\theta \right) \\
1 - \bar{\tau}^s(\theta) &= \frac{1}{\beta R} \frac{U_c(c(\theta), q(\theta), l(\theta), S)}{g(w(\theta), w_2(\theta), \theta, Q)}
\end{aligned}$$

This yields this Hamiltonian, where ν^s is the Lagrange multiplier on the savings wedge constraint.

$$\begin{aligned}
\mathcal{H} &= \left[c + q - D(S)\theta l + \frac{1}{R}V_{t+1} \left(w, w_2, \theta, \int q(\theta) f(\theta) d\theta \right) \right] f \\
&\quad + \psi \left[-U_l(c, q, l, S) \frac{l}{\theta} + \beta w_2 \right] \\
&\quad - \lambda_1 u(\theta) f_t \\
&\quad + \lambda_2 u(\theta) f_2 \\
&\quad + \phi [u - U(c, q, l, S) - \beta w] \\
&\quad - \mu \left[S - S \left(\dots, \int q(\theta) f(\theta) d\theta \right) \right] \\
&\quad + \nu^s \left[1 - \bar{\tau}^s(\theta) - \frac{1}{\beta R} \frac{U_c(c(\theta), q(\theta), l(\theta), S)}{g(w(\theta), w_2(\theta), \theta, Q)} \right]
\end{aligned}$$

The first-order conditions are as follows, where I summarize $g(w(\theta), w_2(\theta), \theta, Q)$ as g for brevity.

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} : f - \psi U_{cl} \frac{l}{\theta} - \nu^s \frac{1}{\beta R} \frac{U_{cc}}{g} &= \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} : f + \frac{1}{R} \frac{\partial V}{\partial Q} f(\theta) - \psi U_{ql} \frac{l}{\theta} + \mu S_2 f - \nu^s \frac{1}{\beta R} \frac{U_{cq}}{g} + \nu^s \frac{1}{\beta R} \frac{U_c}{g^2} g_q f &= \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} : -D(S) \theta f - \phi U_l - \nu^s \frac{1}{\beta R} \frac{U_{cl}}{g} &= \frac{1}{\theta} \psi [U_{ll} l + U_l] \\
\frac{\partial \mathcal{H}}{\partial u} : -\lambda_1 f + \lambda_2 f_2 + \phi &= \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f + \nu^s \frac{1}{\beta R} \frac{U_c}{g^2} g_w f &= \phi \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f + \nu^s \frac{1}{\beta R} \frac{U_c}{g^2} g_{w_2} f &= -\psi \beta \\
\frac{\partial \mathcal{H}}{\partial S} : \int \left(-D'(S) \theta l f - \psi U_{ls} \frac{l}{\theta} - \phi U_S \right) &= \mu
\end{aligned}$$

Through the first-order conditions, one can already see that the constraint on the savings wedge will affect both the labor and commodity wedge. First, I derive the labor wedge to see how it changes more precisely, following the same basic steps. As usual, I first substitute ϕ .

$$\begin{aligned}
& -D(S) \theta f - \frac{f - \psi U_{cl} \frac{l}{\theta} - \nu^s \frac{1}{\beta R} \frac{U_{cc}}{g}}{U_c} U_l - \nu^s \frac{1}{\beta R} \frac{U_{cl}}{g} = \frac{1}{\theta} \psi [U_{ll} l + U_l] \\
\implies & -D(S) \theta f - \frac{U_l}{U_c} f + \psi \frac{U_{cl}}{U_c} \frac{U_{ll}}{\theta} + \nu^s \frac{1}{\beta R} \frac{U_{cc}}{U_c} \frac{U_l}{g} - \nu^s \frac{1}{\beta R} \frac{U_{cl}}{g} = \frac{1}{\theta} \psi [U_{ll} l + U_l] \\
\implies & -D(S) \left(1 + \frac{U_l}{D(S) \theta U_c} \right) \theta f + \psi \frac{U_{cl}}{U_c} \frac{U_{ll}}{\theta} + \nu^s \frac{1}{\beta R} \left[\frac{U_{cc}}{U_c} \frac{U_l}{g} - \frac{U_{cl}}{g} \right] = \frac{1}{\theta} \psi [U_{ll} l + U_l] \\
\implies & -D(S) \tau^y \theta f + \nu^s \frac{1}{\beta R} \left[\frac{U_{cc}}{U_c} \frac{U_l}{g} - \frac{U_{cl}}{g} \right] = -\frac{1}{\theta} \psi (1 + \epsilon) (-U_l) - \psi \frac{U_{cl}}{U_c} \frac{U_{ll}}{\theta} \\
\implies & -D(S) \tau^y \theta f + \nu^s \frac{1}{\beta R} \frac{U_c}{g} \left[\frac{U_{cc}}{U_c} \frac{U_l}{U_c} - \frac{U_{cl}}{U_c} \right] = -\frac{1}{\theta} \psi (1 + \epsilon) (-U_l) - \psi \frac{U_{cl}}{U_c} \frac{U_{ll}}{\theta} \\
\implies & \tau^y - \nu^s (1 - \bar{\tau}^s) \left[\frac{U_{cc}}{U_c} \frac{U_l}{U_c} - \frac{U_{cl}}{U_c} \right] \frac{1}{D(S) \theta f} = -\frac{1}{D(S) \theta^2 f} \psi (1 + \epsilon - \gamma) (-U_l) \\
\implies & \tau^y - \nu^s (1 - \bar{\tau}^s) \left[\frac{U_{cc}}{U_c} - \frac{U_{cl}}{U_l} \right] \frac{U_l}{D(S) \theta U_c f} = -\frac{1}{D(S) \theta^2 f} \psi (1 + \epsilon - \gamma) (-U_l) \\
\implies & \tau^y - \nu^s (1 - \bar{\tau}^s) \left[\frac{U_{cc}}{U_c} - \frac{U_{cl}}{U_l} \right] (1 - \tau^y) = \frac{(1 - \tau^y) U_c}{\theta f} \psi (1 + \epsilon - \gamma) \\
\implies & \frac{\tau^y}{1 - \tau^y} = \frac{\psi U_c}{\theta f} (1 + \epsilon - \gamma) + \nu^s (1 - \bar{\tau}^s) \left[\frac{U_{cc}}{U_c} - \frac{U_{cl}}{U_l} \right]
\end{aligned}$$

I define the following to keep notation short:

$$\Gamma_l^s \equiv \frac{U_{cc}}{U_c} - \frac{U_{cl}}{U_l}$$

If we isolate ψ ,

$$\psi = \left[\left(\frac{\tau^y}{1 - \tau^y} \right) - \nu^s \Gamma_l^s \right] \frac{\theta f}{U_c} \frac{1}{1 + \epsilon - \gamma}$$

Because $\psi = -\frac{1}{\beta R} \frac{\partial V_{t+1}}{\partial w_2} f$,

$$-\frac{\partial V_{t+1}}{\partial w_2} = \beta R \left[\left(\frac{\tau^y}{1 - \tau^y} \right) - \nu^g \Gamma_l^g \right] \frac{\theta}{U_c} \frac{1}{1 + \epsilon - \gamma}$$

By the envelope condition,

$$\lambda_{2,t+1} = \beta R \left[\left(\frac{\tau_{t+1}^y}{1 - \tau_{t+1}^y} \right) - \nu_{t+1}^g \Gamma_{l,t+1}^g \right] \frac{\theta}{U_{c,t+1}} \frac{1}{1 + \epsilon_{t+1} - \gamma_{t+1}}$$

Next, we find the expression for $\psi(\theta)$ by integrating the FOC for u .

$$\begin{aligned} \dot{\psi} &= -\lambda_1 f + \lambda_2 f_2 + \frac{f - \psi U_{cl} \frac{l}{\theta} + \nu^g \left(\frac{U_{cq}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_c} \right)}{U_c} \\ \underbrace{\dot{\psi}}_{\frac{dy}{dx}} &= -\lambda_1 f + \lambda_2 f_2 + \underbrace{\frac{f + \nu^g \left(\frac{U_{cq}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_c} \right)}{U_c}}_Q - \underbrace{\frac{1}{\theta} \gamma}_{P} \underbrace{\psi}_y \end{aligned}$$

This is a first-order differential equation of form $\frac{dy}{dx} = Q + Py$, which has the following solution:

$$\psi(\theta) = \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_c(x)} [1 + \nu^g \Gamma_q^g] f(x) - \lambda_1 f(x) + \lambda_2 f_2(x) \right) dx$$

where

$$\Gamma_q^g \equiv \frac{U_{cq}}{U_cf} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_cf}$$

Using the boundary condition that $\psi(0) = 0$, we can find λ_1 .

$$\lambda_{1,t} = \frac{\int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_{c,t}(x)} [1 + \nu_t^g \Gamma_{q,t}^g] f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx}{\int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_t(x)}$$

We substitute ψ and $\lambda_{2,t}$ (iterating from $\lambda_{2,t+1}$) into the expression for the labor wedge. Since the steps are the same as in Golosov et al. (2016) aside for some different substitutions, I skip some of the algebra.

$$\begin{aligned}
\frac{\tau_t^y}{1 - \tau_t^y} &= \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_{c,t}(x)} [1 + \nu_t^g \Gamma_{q,t}^g] f_t(x) - \lambda_{1,t} f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx \frac{U_{c,t}(\theta)}{\theta_t f_t(\theta)} (1 + \epsilon_t(\theta) - \gamma_t(\theta)) + \nu_t^g \Gamma_{l,t}^g \\
&= \dots \\
&= \int_{\theta}^{\infty} \frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] (1 + \epsilon_t(\theta) - \gamma_t(\theta)) [1 + \nu_t^g \Gamma_{q,t}^g] \frac{f_t(x)}{\theta_t f_t(\theta)} dx - \\
&\quad \int_{\theta}^{\infty} \frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] U_{c,t}(x) (1 + \epsilon_t(\theta) - \gamma_t(\theta)) \lambda_{1,t} \frac{f_t(x)}{\theta_t f_t(\theta)} dx + \\
&\quad \beta R \left[\left(\frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \right) + \nu_t^g \Gamma_{l,t}^g \right] \frac{1 + \epsilon_t(\theta) - \gamma_t(\theta)}{1 + \epsilon_{t-1}(\theta) - \gamma_{t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)} + \nu_t^g \Gamma_{l,t}^g \\
&= \dots \\
&= (1 + \epsilon_t(\theta) - \gamma_t(\theta)) \frac{1 - F_t(\theta)}{\theta_t f_t(\theta)} \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 + \nu_t^g \Gamma_{q,t}^g - \lambda_{1,t} U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} + \\
&\quad \beta R \left[\left(\frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \right) + \nu_t^g \Gamma_{l,t}^g \right] \frac{1 + \epsilon_t(\theta) - \gamma_t(\theta)}{1 + \epsilon_{t-1}(\theta) - \gamma_{t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)} + \nu_t^g \Gamma_{l,t}^g
\end{aligned}$$

This yields the following modified ABCD formula.

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \left[\left(\frac{\tau_t^y}{1 - \tau_t^y} \right) + \nu_t^g \Gamma_{q,t}^g \right] D_t(\theta) + \nu_t^g \Gamma_{l,t}^g$$

where

$$\begin{aligned}
A_t(\theta) &\equiv (1 + \epsilon_t(\theta) - \gamma_t(\theta)) \\
B_t(\theta) &\equiv \frac{1 - F_t(\theta)}{\theta_t f_t(\theta)} \\
C_t(\theta) &\equiv \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 + \nu_t^g \Gamma_{q,t}^g - \lambda_{1,t} U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
D_t(\theta) &\equiv \frac{A_t(\theta)}{A_{t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)}
\end{aligned}$$

C.3 Constrained-optimal labor wedge (suboptimal commodity wedge)

$$\begin{aligned}
V_t(\hat{w}, \hat{w}_2, \theta_-, Q_-) &= \min_{c, q, l, u, w, w_2, S} \int_0^\infty \left[c(\theta) + q(\theta) - D(S)\theta l(\theta) + \frac{1}{R}V_{t+1}(w(\theta), w_2(\theta), \theta, Q) \right] f_t(\theta|\theta_-)d\theta \\
&\quad \text{s.t.} \\
\dot{u}(\theta, S) &= -U_l(c(\theta), q(\theta), l(\theta), S) \frac{1}{\theta} + \beta w_2(\theta) \\
\hat{w} &= \int_0^\infty u(\theta) f_t(\theta|\theta_-)d\theta \\
\hat{w}_2 &= \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_-)d\theta \\
u(\theta, S) &= U(c(\theta), q(\theta), l(\theta), S) + \beta w(\theta) \\
S &= S\left(\dots, \int q(\theta) f(\theta)d\theta\right) \\
1 + \bar{\tau}^g(\theta) &= \frac{U_q(c(\theta), q(\theta), l(\theta), S)}{U_c(c(\theta), q(\theta), l(\theta), S)}
\end{aligned}$$

We get this Hamiltonian, where ν^g is the Lagrange multiplier on the commodity wedge constraint.

$$\begin{aligned}
\mathcal{H} &= \left[c + q - D(S)\theta l + \frac{1}{R}V_{t+1}\left(w, w_2, \theta, \int q(\theta) f(\theta)d\theta\right) \right] f \\
&\quad + \psi \left[-U_l(c, q, l, S) \frac{l}{\theta} + \beta w_2 \right] \\
&\quad - \lambda_1 u(\theta) f_t \\
&\quad + \lambda_2 u(\theta) f_2 \\
&\quad + \phi [u - U(c, q, l, S) - \beta w] \\
&\quad - \mu \left[S - S\left(\dots, \int q(\theta) f(\theta)d\theta\right) \right] \\
&\quad + \nu^g \left[\frac{U_q(c, q, l, S)}{U_c(c, q, l, S)} - 1 - \bar{\tau}^g(\theta) \right]
\end{aligned}$$

The first-order conditions are as follows.

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} + \nu^g \left(\frac{U_{cq}}{U_c} - \frac{U_q U_{cc}}{U_c^2} \right) &= \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f + \frac{1}{R} \frac{\partial V}{\partial Q} f(\theta) - \psi U_{ql} \frac{l}{\theta} + \mu S_2 f + \nu^g \left(\frac{U_{qq}}{U_c} - \frac{U_q U_{cq}}{U_c^2} \right) &= \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -D(S) \theta f - \phi U_l + \nu^g \left(\frac{U_{ql}}{U_c} - \frac{U_q U_{cl}}{U_c^2} \right) &= \frac{1}{\theta} \psi [U_{ll} l + U_l] \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi &= \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f &= \phi \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f &= -\psi \beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int \left(-D'(S) \theta l f - \psi U_{ls} \frac{l}{\theta} - \phi U_S + \nu^g \left(\frac{U_{qS}}{U_c} - \frac{U_q U_{cS}}{U_c^2} \right) \right) &= \mu
\end{aligned}$$

Using the commodity wedge constraint $1 + \bar{\tau}^g(\theta) = \frac{U_q}{U_c}$,

$$\begin{aligned}
\frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} + \nu^g \left(\frac{U_{cq}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_c} \right) &= \phi U_c \\
\frac{\partial \mathcal{H}}{\partial q} &: f + \frac{1}{R} \frac{\partial V}{\partial Q} f(\theta) - \psi U_{ql} \frac{l}{\theta} + \mu S_2 f + \nu^g \left(\frac{U_{qq}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cq}}{U_c} \right) &= \phi U_q \\
\frac{\partial \mathcal{H}}{\partial l} &: -D(S) \theta f - \phi U_l + \nu^g \left(\frac{U_{ql}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cl}}{U_c} \right) &= \frac{1}{\theta} \psi [U_{ll} l + U_l] \\
\frac{\partial \mathcal{H}}{\partial u} &: -\lambda_1 f + \lambda_2 f_2 + \phi &= \dot{\psi} \\
\frac{\partial \mathcal{H}}{\partial w} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f &= \phi \beta \\
\frac{\partial \mathcal{H}}{\partial w_2} &: \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f &= -\psi \beta \\
\frac{\partial \mathcal{H}}{\partial S} &: \int \left(-D'(S) \theta l f - \psi U_{ls} \frac{l}{\theta} - \phi U_S + \nu^g \left(\frac{U_{qS}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cS}}{U_c} \right) \right) &= \mu
\end{aligned}$$

Next, we substitute ϕ using the first FOC into the other FOCs.

$$\phi = \frac{f - \psi U_{cl} \frac{l}{\theta} + \nu^g \left(\frac{U_{cq}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_c} \right)}{U_c}$$

Let's start deriving the expression for the (constrained)-optimal labor wedge, taking $\frac{\partial \mathcal{H}}{\partial l}$ as a starting point and using $1 - \tau^y = -\frac{U_l}{D(S)\theta U_c}$.

$$\begin{aligned} & -D(S)\theta f - \frac{f - \psi U_{cl} \frac{l}{\theta} + \nu^g \left(\frac{U_{cq}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_c} \right)}{U_c} U_l + \nu^g \left(\frac{U_{ql}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cl}}{U_c} \right) = \frac{1}{\theta} \psi [U_{ll} + U_l] \\ \implies & -D(S)\theta f - \frac{U_l}{U_c} f + \psi \frac{U_{cl}}{U_c} \frac{U_{ll}}{\theta} - \nu^g \left(\frac{U_{cq}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_c} \right) \frac{U_l}{U_c} + \nu^g \left(\frac{U_{ql}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cl}}{U_c} \right) = \frac{1}{\theta} \psi [U_{ll} + U_l] \\ \implies & -D(S)\theta f - \frac{U_l}{U_c} f + \psi \frac{U_{cl}}{U_c} \frac{U_{ll}}{\theta} - \nu^g \left(\frac{U_{cq}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_c} \right) \frac{U_l}{U_c} + \nu^g \left(\frac{U_{ql}}{U_l} - (1 + \bar{\tau}^g) \frac{U_{cl}}{U_l} \right) \frac{U_l}{U_c} = \frac{1}{\theta} \psi [U_{ll} + U_l] \\ \implies & -D(S)\theta f - \frac{U_l}{U_c} f + \psi \frac{U_{cl}}{U_c} \frac{U_{ll}}{\theta} - \nu^g \left(\frac{U_{cq}}{U_c} - \frac{U_{ql}}{U_l} - (1 + \bar{\tau}^g) \left(\frac{U_{cc}}{U_c} - \frac{U_{cl}}{U_l} \right) \right) \frac{U_l}{U_c} = \frac{1}{\theta} \psi [U_{ll} + U_l] \\ \implies & -D(S) \left(1 + \frac{U_l}{D(S)\theta U_c} \right) \theta f + \psi \frac{U_{cl}}{U_c} \frac{U_{ll}}{\theta} - \nu^g \left(\frac{U_{cq}}{U_c} - \frac{U_{ql}}{U_l} - (1 + \bar{\tau}^g) \left(\frac{U_{cc}}{U_c} - \frac{U_{cl}}{U_l} \right) \right) \frac{U_l}{U_c} = \frac{1}{\theta} \psi [U_{ll} + U_l] \\ \implies & -D(S)\tau^y \theta f + \psi \frac{U_{cl}}{U_c} \frac{U_{ll}}{\theta} - \nu^g \left(\frac{U_{cq}}{U_c} - \frac{U_{ql}}{U_l} - (1 + \bar{\tau}^g) \left(\frac{U_{cc}}{U_c} - \frac{U_{cl}}{U_l} \right) \right) \frac{U_l}{U_c} = -\frac{1}{\theta} \psi \left[\frac{U_{ll} + U_l}{U_l} \right] (-U_l) \\ \implies & -D(S)\tau^y \theta f - \nu^g \left(\frac{U_{cq}}{U_c} - \frac{U_{ql}}{U_l} - (1 + \bar{\tau}^g) \left(\frac{U_{cc}}{U_c} - \frac{U_{cl}}{U_l} \right) \right) \frac{U_l}{U_c} = -\frac{1}{\theta} \psi (1 + \epsilon) (-U_l) - \psi \frac{U_{cl}}{U_c} \frac{U_{ll}}{\theta} \\ \implies & \tau^y + \nu^g \left(\frac{U_{cq}}{U_c} - \frac{U_{ql}}{U_l} - (1 + \bar{\tau}^g) \left(\frac{U_{cc}}{U_c} - \frac{U_{cl}}{U_l} \right) \right) \frac{U_l}{D(S)\theta U_c} \frac{1}{f} = -\frac{1}{D(S)\theta^2 f} \psi (1 + \epsilon - \gamma) (-U_l) \\ \implies & \tau^y - \nu^g \left(\frac{U_{cq}}{U_c f} - \frac{U_{ql}}{U_l f} - (1 + \bar{\tau}^g) \left(\frac{U_{cc}}{U_c f} - \frac{U_{cl}}{U_l f} \right) \right) (1 - \tau^y) = \frac{(1 - \tau^y) U_c}{\theta f} \psi (1 + \epsilon - \gamma) \\ \implies & \frac{\tau^y}{1 - \tau^y} = \frac{U_c}{\theta f} \psi (1 + \epsilon - \gamma) + \nu^g \left(\frac{U_{cq}}{U_c f} - \frac{U_{ql}}{U_l f} - (1 + \bar{\tau}^g) \left(\frac{U_{cc}}{U_c f} - \frac{U_{cl}}{U_l f} \right) \right) \end{aligned}$$

I define the following to keep notation short:

$$\Gamma_l^g \equiv \frac{U_{cq}}{U_c f} - \frac{U_{ql}}{U_l f} - (1 + \bar{\tau}^g) \left(\frac{U_{cc}}{U_c f} - \frac{U_{cl}}{U_l f} \right)$$

If we isolate ψ ,

$$\psi = \left[\left(\frac{\tau^y}{1 - \tau^y} \right) - \nu^g \Gamma_l^g \right] \frac{\theta f}{U_c} \frac{1}{1 + \epsilon - \gamma}$$

Because $\psi = -\frac{1}{\beta R} \frac{\partial V_{t+1}}{\partial w_2} f$,

$$-\frac{\partial V_{t+1}}{\partial w_2} = \beta R \left[\left(\frac{\tau^y}{1 - \tau^y} \right) - \nu^g \Gamma_l^g \right] \frac{\theta}{U_c} \frac{1}{1 + \epsilon - \gamma}$$

By the envelope condition,

$$\lambda_{2,t+1} = \beta R \left[\left(\frac{\tau_{t+1}^y}{1 - \tau_{t+1}^y} \right) - \nu_{t+1}^g \Gamma_{l,t+1}^g \right] \frac{\theta}{U_{c,t+1}} \frac{1}{1 + \epsilon_{t+1} - \gamma_{t+1}}$$

Next, we find the expression for $\psi(\theta)$ by integrating the FOC for u .

$$\begin{aligned} \dot{\psi} &= -\lambda_1 f + \lambda_2 f_2 + \frac{f - \psi U_{cl} \frac{l}{\theta} + \nu^g \left(\frac{U_{cq}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_c} \right)}{U_c} \\ \underbrace{\frac{d\psi}{dx}}_{\frac{dy}{dx}} &= -\lambda_1 f + \lambda_2 f_2 + \underbrace{\frac{f + \nu^g \left(\frac{U_{cq}}{U_c} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_c} \right)}{U_c}}_Q - \underbrace{\frac{1}{\theta} \gamma \psi}_{Py} \end{aligned}$$

This is a first-order differential equation of form $\frac{dy}{dx} = Q + Py$, which has the following solution:

$$\psi(\theta) = \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_c(x)} [1 + \nu^g \Gamma_q^g] f(x) - \lambda_1 f(x) + \lambda_2 f_2(x) \right) dx$$

where

$$\Gamma_q^g \equiv \frac{U_{cq}}{U_c f} - (1 + \bar{\tau}^g) \frac{U_{cc}}{U_c f}$$

Using the boundary condition that $\psi(0) = 0$, we can find λ_1 .

$$\lambda_{1,t} = \frac{\int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_{c,t}(x)} [1 + \nu_t^g \Gamma_{q,t}^g] f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx}{\int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_t(x)}$$

We substitute ψ and $\lambda_{2,t}$ (iterating from $\lambda_{2,t+1}$) into the expression for the labor wedge. Since the steps are the same as in Golosov et al. (2016) aside for some different substitutions, I skip some of the algebra.

$$\begin{aligned}
\frac{\tau_t^y}{1 - \tau_t^y} &= \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_{c,t}(x)} [1 + \nu_t^g \Gamma_{q,t}^g] f_t(x) - \lambda_{1,t} f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx \frac{U_{c,t}(\theta)}{\theta_t f_t(\theta)} (1 + \epsilon_t(\theta) - \gamma_t(\theta)) + \nu_t^g \Gamma_{l,t}^g \\
&= \dots \\
&= \int_{\theta}^{\infty} \frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] (1 + \epsilon_t(\theta) - \gamma_t(\theta)) [1 + \nu_t^g \Gamma_{q,t}^g] \frac{f_t(x)}{\theta_t f_t(\theta)} dx - \\
&\quad \int_{\theta}^{\infty} \frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] U_{c,t}(x) (1 + \epsilon_t(\theta) - \gamma_t(\theta)) \lambda_{1,t} \frac{f_t(x)}{\theta_t f_t(\theta)} dx + \\
&\quad \beta R \left[\left(\frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \right) + \nu_t^g \Gamma_{l,t}^g \right] \frac{1 + \epsilon_t(\theta) - \gamma_t(\theta)}{1 + \epsilon_{t-1}(\theta) - \gamma_{t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)} + \nu_t^g \Gamma_{l,t}^g \\
&= \dots \\
&= (1 + \epsilon_t(\theta) - \gamma_t(\theta)) \frac{1 - F_t(\theta)}{\theta_t f_t(\theta)} \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 + \nu_t^g \Gamma_{q,t}^g - \lambda_{1,t} U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} + \\
&\quad \beta R \left[\left(\frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \right) + \nu_t^g \Gamma_{l,t}^g \right] \frac{1 + \epsilon_t(\theta) - \gamma_t(\theta)}{1 + \epsilon_{t-1}(\theta) - \gamma_{t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)} + \nu_t^g \Gamma_{l,t}^g
\end{aligned}$$

This yields the following modified ABCD formula.

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \left[\left(\frac{\tau_t^y}{1 - \tau_t^y} \right) + \nu_t^g \Gamma_{q,t}^g \right] D_t(\theta) + \nu_t^g \Gamma_{l,t}^g$$

where

$$\begin{aligned}
A_t(\theta) &\equiv (1 + \epsilon_t(\theta) - \gamma_t(\theta)) \\
B_t(\theta) &\equiv \frac{1 - F_t(\theta)}{\theta_t f_t(\theta)} \\
C_t(\theta) &\equiv \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} - \gamma_t(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 + \nu_t^g \Gamma_{q,t}^g - \lambda_{1,t} U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
D_t(\theta) &\equiv \frac{A_t(\theta)}{A_{t-1}(\theta)} \frac{U_{c,t}(\theta)}{U_{c,t-1}(\theta)} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta_t f_t(\theta)}
\end{aligned}$$

C.4 Constrained-optimal savings wedge (suboptimal commodity wedge)

TBA